THE p-PRIMARY PART OF THE TATE–SHAFAREVICH GROUP IN RANKS THREE TO SIX

DEEP BHATTACHARJEE

Abstract. We prove that $Ш(E/\mathbf{Q})[p] = 0$ for the rank-four curve 234446a1 at every prime $5 \le p < 1000$ except 337, for the rank-five and rank-six curves of smallest known conductor at small primes, and for the rank-three curves of conductor between 30 000 and 100 000 at every good ordinary prime $5 \le p < 100$.

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MSC 2020: Primary 11G40; Secondary 11G05, 11R23, 11F67, 11G50.

1. Introduction

1.1. **The conjecture.** Let E be an elliptic curve over $\mathbf{Q}$ of conductor N. By the theorem of Mordell the group $E(\mathbf{Q})$ is finitely generated, and by the modularity theorem [39, 1] the Hasse–Weil function $L(E, s)$ is the L-function of a newform of weight two and level N, so it extends to an entire function with a functional equation relating s and $2 - s$. The conjecture of Birch and Swinnerton-Dyer [7] asserts that

$$\text{(1)}\quad \operatorname{ord}_{s=1} L(E,s) = r := \operatorname{rank} E(\mathbf{Q}), \qquad \frac{L^{(r)}(E,1)}{r!\,\Omega_E} = \frac{\prod_v c_v \cdot \#Ш(E/\mathbf{Q}) \cdot \operatorname{Reg}(E/\mathbf{Q})}{\#E(\mathbf{Q})^2_{\mathrm{tors}}},$$

where Ω_E is the real period of a Néron differential, c_v are the Tamagawa numbers, Reg is the regulator of the Néron–Tate height, and $Ш(E/\mathbf{Q})$ is the Tate–Shafarevich group, which the second statement presupposes to be finite.

When $\operatorname{ord}_{s=1} L(E,s) \le 1$ the first statement and the finiteness of Ш are theorems, through the work of Gross and Zagier [15] and Kolyvagin [19]; much of the leading-term formula is known as well, in large part through the main conjecture of Iwasawa theory proved by Kato [16] and Skinner and Urban [37]. As soon as the order of vanishing is at least two, nothing of this kind is known: for no elliptic curve over $\mathbf{Q}$ of rank at least two has $Ш(E/\mathbf{Q})$ been proved finite.

1.2. **One prime at a time.** The p-primary part $Ш(E/\mathbf{Q})[p^\infty]$ behaves better. Mazur's control theorem [21] ties it to the Selmer group of E over the cyclotomic $\mathbf{Z}_p$-extension of $\mathbf{Q}$, and two deep theorems pin down that Selmer group from both sides. Kato's Euler system shows that its characteristic series $f_E(T)$ divides the p-adic L-function $L_p(E,T)$ of Mazur, Tate and Teitelbaum [27]; Schneider [32, 33] and Perrin-Riou [30] showed that f_E vanishes at $T = 0$ to order at least $\operatorname{rank} E(\mathbf{Q})$, and they computed the leading coefficient in terms of $\#Ш(E/\mathbf{Q})[p^\infty]$ and a p-adic regulator. The first theorem bounds f_E from above, the second from below, and the gap between them is measured by a single coefficient of $L_p(E,T)$, which can be computed exactly from modular symbols.

Stein and Wuthrich [38] turned this into an algorithm and proved $Ш(E/\mathbf{Q})[p] = 0$ for the 1 534 422 pairs (E,p) in which E has no complex multiplication, has rank at least two and conductor at most 30 000, p is a good ordinary prime with $5 \le p < 1000$, and the mod-p

representation of E is surjective. Every curve in that range has rank two or three; the first curve of rank four has conductor 234 446.

1.3. **Results.** Four theorems are proved here. The first concerns the curve

$$E_4\colon\ y^2+xy=x^3-x^2-79x+289$$

of conductor $234\,446 = 2\cdot 117\,223$, the curve labelled $234446a1$ by Cremona [9]. It is the only curve of rank four of conductor below 500 000.

Theorem A. *The group $E_4(\mathbf{Q})$ is free of rank four; the points $(6,-1)$, $(4,3)$, $(5,-2)$, $(8,7)$ generate a subgroup of index prime to every prime $5\le p<1000$. For every prime p with $5\le p<1000$ and $p\neq 337$,*

$$Ш(E_4/\mathbf{Q})[p^\infty]=0.$$

At each of these primes other than the supersingular primes 59 and 101, the p-adic L-function $L_p(E_4,T)$ vanishes at $T=0$ to order exactly four, the cyclotomic p-adic height pairing on $E_4(\mathbf{Q})$ is nondegenerate, and $L_p(E_4,T)$ generates the characteristic ideal of the dual Selmer group over the cyclotomic $\mathbf{Z}_p$-extension. Moreover $Ш(E_4/\mathbf{Q})[2]=0$.

The second theorem extends the range of Stein and Wuthrich in conductor, for the curves of rank three.

Theorem B. *Let $E/\mathbf{Q}$ be one of the 505 elliptic curves of conductor $30\,000<N\le 100\,000$ whose Mordell–Weil rank is listed as three in Cremona's tables; they form 494 isogeny classes. Then $\operatorname{rank}E(\mathbf{Q})=3$, and for every good ordinary prime $5\le p<100$ of E,*

$$Ш(E/\mathbf{Q})[p^\infty]=0,$$

$\operatorname{ord}_{T=0}L_p(E,T)=3$, and the cyclotomic p-adic height pairing on $E(\mathbf{Q})$ is nondegenerate. This covers 10 404 pairs (E,p), counting one curve in each isogeny class.

The third and fourth theorems concern the curves of rank five and six of smallest known conductor, found by Elkies and Watkins [13]:

$$E_5\colon\ y^2+y=x^3-79x+342,\qquad E_6\colon\ y^2+xy=x^3+x^2-2582x+48720,$$

of conductors 19 047 851 (a prime) and $5\,187\,563\,742 = 2\cdot 3\cdot 2777\cdot 311\,341$.

Theorem C. *$E_5(\mathbf{Q})$ is free of rank five, and $Ш(E_5/\mathbf{Q})[p^\infty]=0$ for $p\in\{7,11,13,17,19\}$. At these primes $L_p(E_5,T)$ vanishes to order exactly five and the p-adic height pairing is nondegenerate.*

Theorem D. *$E_6(\mathbf{Q})$ is free of rank six, and $Ш(E_6/\mathbf{Q})[7^\infty]=0$. The 7-adic L-function of E_6 vanishes to order exactly six and the 7-adic height pairing is nondegenerate.*

The ranks in Theorems A–D are not taken from a descent: they come out of the same argument, which bounds the rank from above by the order of vanishing of $L_p(E,T)$.[1] The nondegeneracy of the p-adic height pairing is Schneider's conjecture [32] at the pair (E,p), and the statement $\operatorname{ord}_T L_p(E,T)=\operatorname{rank}E(\mathbf{Q})$ is the first half of the p-adic Birch–Swinnerton-Dyer conjecture of Mazur, Tate and Teitelbaum [27]. Neither is known in general; both are proved here for every pair covered by the theorems.

The numbers compared in the proof agree beyond what the theorems need. For every pair of Theorem B at $p\le 13$ the coefficient a_3 of $L_p(E,T)$ and the p-adic regulator term $S_p(E)$

[1] The lower bound is given by explicit points. The upper bound $\operatorname{rank}E(\mathbf{Q})\le\operatorname{ord}_T L_p(E,T)$ holds for every good ordinary $p\ge 5$ with surjective mod-p representation, by Theorems 4.1 and 4.2; one coefficient of L_p therefore bounds the rank.

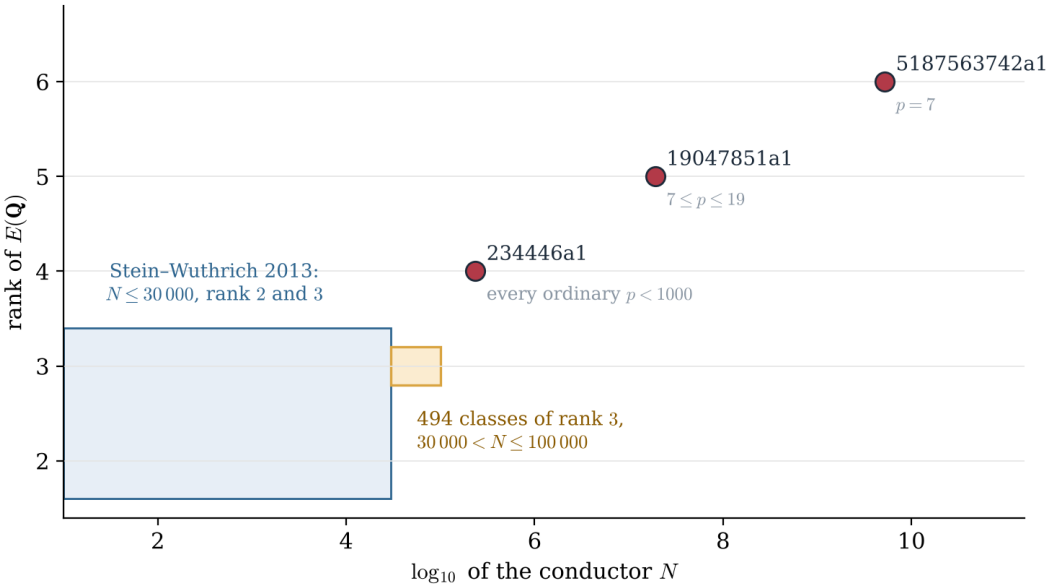


FIGURE 1. Ranks and conductors covered. The shaded rectangle is the range of Stein and Wuthrich; the small box is Theorem B; the three dots are Theorems A, C, D, with the primes treated.

agree in their first two p-adic digits (Proposition 11.1). The analytic order of Ш, computed numerically, is 1 for all these curves, so this is the p-adic Birch–Swinnerton-Dyer formula itself modulo p^{v+2}, with the order of Ш that the classical conjecture predicts.

Figure 1 places the results beside the range of [38].

1.4. **What the method does not reach.** The theorems are statements about one prime at a time. They do not show that $Ш(E/\mathbf{Q})$ is finite for any of these curves: the primes 2 and 3 for E_5 and E_6, the prime 3 for E_4, the primes 2, 3 and the primes of bad or supersingular reduction for the curves of Theorem B, and all primes above the bounds remain. Nor do they touch the complex side of (1). For E_4 the sign of the functional equation is $+1$ and the modular symbol $[0]^+$ vanishes, so $L(E_4, 1) = L'(E_4, 1) = 0$; numerically $L(E_4, s)$ vanishes to order four at $s = 1$, but no method is known that proves $L''(E_4, 1) = 0$, and so the first statement of (1) is open for E_4, as it is for every curve of rank at least four. Section 12 describes what would be needed to pass from these p-adic statements to the conjecture itself.

1.5. **The argument and the plan.** The proof of each case has the same shape, drawn in Figure 2. Sections 2–4 recall the p-adic L-function, the p-adic height and the two theorems of Iwasawa theory on which everything rests, and prove the elementary facts about them that the argument uses: integrality of the modular symbols, the distribution relation, and a bound for the error of the Riemann sums that approximate $L_p(E, T)$. Section 5 proves the criterion (Theorem 5.1) that turns one congruence between two explicit numbers into the conclusions above. Sections 6 and 7 give the two side conditions—surjectivity of $\bar{\rho}_{E,p}$ and p-saturation of the points—in a form that can be checked by counting points on reductions. Sections 8–10 prove Theorems A–D; for E_4 at $p = 5$ and for E_5 at $p = 7$ the whole verification is carried out in the text from a handful of modular symbols. Section 11 compares the leading coefficients beyond their valuations, and Section 12 discusses the distance to the conjecture.

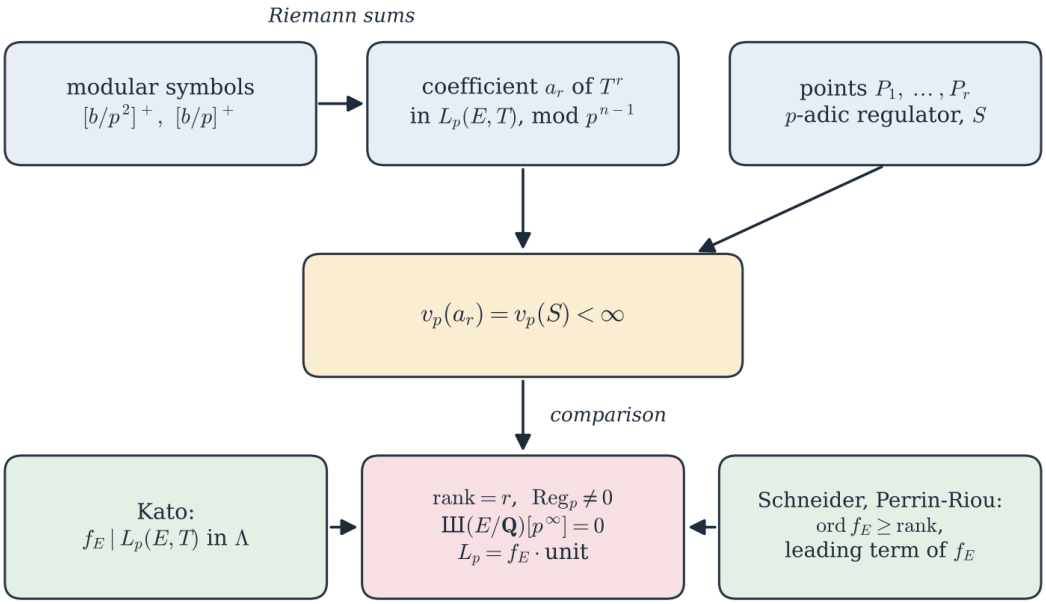


FIGURE 2. The structure of the proof of Theorem 5.1. The top row is computed; the bottom corners are the two theorems of Iwasawa theory used.

The appendices contain the proofs of the auxiliary lemmas, the subgroups of $\mathrm{GL}_2(\mathbf{F}_p)$, the p-adic sigma function in closed form, and the tables.

Finite certificates of this kind—explicit data that reduce a theorem to arithmetic anyone can redo—accompany several recent computations on character sums, totients and modular forms [5, 4, 6].

Notation. Throughout, p is an odd prime, v_p the p-adic valuation on $\mathbf{Q}_p$ with $v_p(0) = \infty$, $\log_p$ the Iwasawa logarithm, and $\Lambda = \mathbf{Z}_p[[T]]$. For an elliptic curve $E/\mathbf{Q}$ we write a_ℓ for the trace of Frobenius at a good prime ℓ, so that $\#\tilde{E}(\mathbf{F}_\ell) = \ell + 1 - a_\ell$, and $\bar{\rho}_{E,p}\colon \mathrm{Gal}(\bar{\mathbf{Q}}/\mathbf{Q}) \to \mathrm{GL}_2(\mathbf{F}_p)$ for the action on $E[p]$. The prime p is *good ordinary* if $p \nmid N$ and $p \nmid a_p$; then $X^2 - a_p X + p$ has exactly one root $\alpha \in \mathbf{Z}_p^\times$.

2. Modular symbols and the p-adic L-function

2.1. **Modular symbols.** Let $f = \sum a_n q^n$ be the newform attached to E. For $r \in \mathbf{Q}$ put $\lambda(r) = 2\pi i \int_r^{i\infty} f(\tau)\, d\tau$, the integral taken along a vertical line, and

$$\lambda^+(r) = -\tfrac{1}{2}\big(\lambda(r) + \lambda(-r)\big), \qquad [r]^+ = \frac{\lambda^+(r)}{\Omega_E}.$$

Since f has real coefficients, $\lambda(-r) = \overline{\lambda(r)}$ and $\lambda^+(r)$ is real. The function $r \mapsto [r]^+$ is the *plus modular symbol* of E; at $r = 0$ it gives $[0]^+ = L(E,1)/\Omega_E$.[2] The following three properties are all that is used.

Lemma 2.1. *Let $r \in \mathbf{Q}$.*

(a) $[r+1]^+ = [r]^+$ *and* $[-r]^+ = [r]^+$.

(b) *For every prime $\ell \nmid N$, $a_\ell\,[r]^+ = \displaystyle\sum_{b=0}^{\ell-1} \Big[\frac{r+b}{\ell}\Big]^+ + [\ell r]^+$.*

(c) *Let p be an odd prime with $p \nmid N$ and $E[p]$ irreducible as a Galois module. If the denominator of r is prime to N, then $[r]^+ \in \mathbf{Z}_{(p)}$.*

Parts (a) and (b) are proved in Appendix B. Part (c) rests on the theorem of Manin and Drinfeld [20, 11] and on Edixhoven's theorem that the Manin constant of an optimal curve is an integer [12]; since the argument is short we give it here.

Proof of (c). We first show that every isogeny ψ between E and a curve E' isogenous to it can be chosen of degree prime to p. Any isogeny factors as a multiplication map followed by an isogeny with cyclic kernel, so it is enough to consider cyclic kernels C. If p divided $\#C$, then C would contain a unique subgroup of order p; it would be stable under $\mathrm{Gal}(\bar{\mathbf{Q}}/\mathbf{Q})$, because C is, and it would be a line in $E[p]$ fixed by Galois, contradicting irreducibility. Hence $p \nmid \#C$.

Let E_0 be the optimal curve in the isogeny class of E, with modular parametrisation $\pi_0\colon X_0(N) \to E_0$ and Manin constant c_0, so that $c_0\,\pi_0^*\omega_{E_0} = 2\pi i f(\tau)\,d\tau$ for the Néron differential ω_{E_0}; by [12], $c_0 \in \mathbf{Z}$. Choose $\psi\colon E_0 \to E$ of degree d prime to p and put $\pi = \psi \circ \pi_0$. Néron differentials pull back to integral multiples under isogenies, so $\psi^*\omega_E = m\,\omega_{E_0}$ and $\hat\psi^*\omega_{E_0} = m'\omega_E$ with $m, m' \in \mathbf{Z}$; since $\psi \circ \hat\psi$ is multiplication by d we get $mm' = d$, hence $p \nmid m$. Thus $c\,\pi^*\omega_E = 2\pi i f\,d\tau$ with $c = c_0/m \in \mathbf{Z}_{(p)}$.

Write $r = a/s$ in lowest terms with $\gcd(s, N) = 1$. Since $\gcd(s, aN) = 1$ there are integers x, y with $xs - yaN = 1$. The matrix $\left(\begin{smallmatrix} x & a \\ yN & s \end{smallmatrix}\right)$ then lies in $\Gamma_0(N)$, and it sends 0 to $a/s = r$. So r and 0 are the same cusp of $X_0(N)$, and a path from r to 0 in the upper half-plane maps to a closed loop on $X_0(N)$. Therefore

$$\lambda(r) - \lambda(0) = 2\pi i \int_r^0 f(\tau)\,d\tau = c \int_{\pi(\text{loop})} \omega_E \in c\,\Lambda_E,$$

where Λ_E is the period lattice of ω_E. Next, $\lambda(0) = c\int_{\pi(0)}^{\pi(\infty)} \omega_E$ is an elliptic logarithm of the point $Q = \pi(\infty) - \pi(0) \in E$. The cusps 0 and ∞ are rational points of $X_0(N)$, so $Q \in E(\mathbf{Q})$, and Q is a torsion point by the theorem of Manin and Drinfeld. Its order t divides $\#E(\mathbf{Q})_{\mathrm{tors}}$, which is prime to p because $E(\mathbf{Q})[p]$ is a Galois-stable subspace of the irreducible module $E[p]$ and is therefore 0. Hence $\lambda(\pm r) \in (c/t)\Lambda_E$ and $\lambda^+(r) \in \frac{c}{2t}\Lambda_E \cap \mathbf{R} = \frac{c}{2t}\mathbf{Z}\,\Omega_0$, where Ω_0 is the least positive real period. Finally $\Omega_E = c_\infty\Omega_0$ with $c_\infty \in \{1, 2\}$ the number of connected components of $E(\mathbf{R})$, so $[r]^+ \in \frac{c}{2tc_\infty}\mathbf{Z} \subset \mathbf{Z}_{(p)}$. □

[2] With the orientation used here $\lambda(0) = 2\pi i \int_0^{i\infty} f = -L(E,1)$, the integral $\int_0^{i\infty} f(\tau)\,d\tau$ being $iL(E,1)/(2\pi)$; the sign in the definition of λ^+ compensates. The proofs use only p-adic valuations, which a sign or a factor 2 does not change. The congruences between a_r and S_p reported in §§8–10 do depend on the normalisation, and they hold for the one fixed here: division by the Néron period Ω_E, which is twice the least positive real period when $E(\mathbf{R})$ has two components.

2.2. **How the symbols are evaluated.** The values $[r]^+$ are rational numbers, and they can be found exactly by a finite computation. The device is Manin's: a path from ∞ to a rational number is cut into unimodular pieces by the continued fraction of that number, and each piece is carried by an element of $\mathrm{SL}_2(\mathbf{Z})$ to the path from 0 to ∞.

Let $r = a/s$ with $s \geq 1$ and $\gcd(a, s) = 1$, and let p_k/q_k for $0 \leq k \leq m$ be the convergents of its continued fraction, with $p_m/q_m = a/s$, and $p_{-1}/q_{-1} = 1/0$. They satisfy $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$, so

$$g_k = \begin{pmatrix} p_k & (-1)^{k-1} p_{k-1} \\ q_k & (-1)^{k-1} q_{k-1} \end{pmatrix} \in \mathrm{SL}_2(\mathbf{Z}), \qquad g_k \cdot 0 = \frac{p_{k-1}}{q_{k-1}}, \quad g_k \cdot \infty = \frac{p_k}{q_k}.$$

Write $\{\alpha, \beta\}$ for the path from α to β in the upper half-plane together with the cusps. Then $\{\infty, r\} = \sum_{k=0}^{m} \{g_k 0, g_k \infty\}$, and therefore

$$\lambda(r) = -\sum_{k=0}^{m} \xi\big(q_k : (-1)^{k-1} q_{k-1}\big), \qquad \xi(c : d) = 2\pi i \int_{g \cdot 0}^{g \cdot \infty} f(\tau)\, d\tau, \tag{2}$$

where $g \in \mathrm{SL}_2(\mathbf{Z})$ is any matrix with bottom row (c, d). The integral in the definition of ξ depends only on the coset $\Gamma_0(N) g$, because $f(\gamma\tau)\, d(\gamma\tau) = f(\tau)\, d\tau$ for $\gamma \in \Gamma_0(N)$, and that coset is determined by the point $(c : d)$ of $\mathbf{P}^1(\mathbf{Z}/N\mathbf{Z})$. So ξ is a function on the finite set $\mathbf{P}^1(\mathbf{Z}/N\mathbf{Z})$, of cardinality $N \prod_{\ell \mid N} (1 + \ell^{-1})$; its values are the *Manin symbols* of f.

Two consequences matter here. First, the number of terms in (2) is the length of the continued fraction, which is $O(\log s)$; a symbol with denominator p^n costs $O(n \log p)$ table look-ups once the Manin symbols are known. Second, the Manin symbols satisfy the relations $\xi(x) + \xi(xS) = 0$ and $\xi(x) + \xi(xU) + \xi(xU^2) = 0$, with $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $U = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ (they come from the paths $\{0, \infty\} + \{\infty, 0\} = 0$ and $\{0, \infty\} + \{\infty, 1\} + \{1, 0\} = 0$), and they are eigenvectors of the Hecke operators. Cremona's method [9, Ch. 2] solves this finite system of linear equations over $\mathbf{Q}$ and isolates the eigenvector of f; the result, scaled by the period, gives the exact values of $[r]^+$. This is how the symbols for the curves of conductor below 250 000 were computed. For the two curves of rank five and six the space is too large, and Section 10 uses a numerical method with a proven error bound instead.

2.3. **The measure.** Let $p \nmid N$ be good ordinary, with unit root α. For $k \geq 1$ and $a \in \mathbf{Z}$ prime to p put

$$\mu(a + p^k \mathbf{Z}_p) = \frac{1}{\alpha^k} \Big[\frac{a}{p^k}\Big]^+ - \frac{1}{\alpha^{k+1}} \Big[\frac{a}{p^{k-1}}\Big]^+. \tag{3}$$

By Lemma 2.1(a) the right side depends only on a modulo p^k.

Lemma 2.2. *The function μ is additive on the compact open subsets of $\mathbf{Z}_p^\times$: for $k \geq 1$ and a prime to p, $\sum_{b=0}^{p-1} \mu(a + bp^k + p^{k+1}\mathbf{Z}_p) = \mu(a + p^k \mathbf{Z}_p)$. If $E[p]$ is irreducible, μ takes values in $\mathbf{Z}_p$.*

Proof. By Lemma 2.1(a), $[(a + bp^k)/p^k]^+ = [a/p^k]^+$, so

$$\sum_b \mu(a + bp^k + p^{k+1}\mathbf{Z}_p) = \alpha^{-k-1} \sum_b \Big[\frac{a/p^k + b}{p}\Big]^+ - \alpha^{-k-2}\, p \Big[\frac{a}{p^k}\Big]^+.$$

Lemma 2.1(b) with $\ell = p$ and $r = a/p^k$ turns the first sum into $a_p [a/p^k]^+ - [a/p^{k-1}]^+$. Write $\beta = p/\alpha$, the other root, so that $a_p - p/\alpha = \alpha$. The total becomes $\alpha^{-k-1}\big(\alpha [a/p^k]^+ - [a/p^{k-1}]^+\big) = \mu(a + p^k \mathbf{Z}_p)$. Integrality follows from Lemma 2.1(c), as $\alpha \in \mathbf{Z}_p^\times$ and the denominators p^k are prime to N. □

A finitely additive $\mathbf{Z}_p$-valued function on the compact open sets of $\mathbf{Z}_p^\times$ is a measure, and continuous functions $\mathbf{Z}_p^\times \to \mathbf{Z}_p$ can be integrated against it as limits of Riemann sums.

2.4. **The p-adic L-function.** Every $x \in \mathbf{Z}_p^\times$ factors uniquely as $x = \omega(x)\langle x\rangle$ with $\omega(x)^{p-1} = 1$ and $\langle x\rangle \in 1+p\mathbf{Z}_p$. The logarithm maps $1+p\mathbf{Z}_p$ isomorphically onto $p\mathbf{Z}_p$, and $v_p(\log_p(1+p)) = 1$, so

$$\ell(x) = \frac{\log_p\langle x\rangle}{\log_p(1+p)} \in \mathbf{Z}_p$$

and $\langle x\rangle = (1+p)^{\ell(x)}$. Following [27] and [38, §3] we define

$$L_p(E,T) = \int_{\mathbf{Z}_p^\times} (1+T)^{\ell(x)}\, d\mu(x) = \sum_{k\ge 0} a_k T^k, \qquad a_k = \int_{\mathbf{Z}_p^\times} \binom{\ell(x)}{k} d\mu(x). \tag{4}$$

Since $s \mapsto \binom{s}{k}$ maps $\mathbf{Z}_p$ to $\mathbf{Z}_p$ continuously and μ is $\mathbf{Z}_p$-valued, every a_k lies in $\mathbf{Z}_p$: $L_p(E,T) \in \Lambda$ *whenever $E[p]$ is irreducible.* At $T = 0$ the integral is the total mass, $a_0 = (1-\alpha^{-1})^2[0]^+$ (from (3) with $k = 1$ and Lemma 2.1(b)), which is the interpolation property at the trivial character.

The coefficients are computed by Riemann sums. As a runs through $1,\dots,p-1$ and j through $0,\dots,p^{n-1}-1$, the numbers $\omega(a)(1+p)^j$ run once through a set of representatives of $(\mathbf{Z}/p^n\mathbf{Z})^\times$, and $\ell(\omega(a)(1+p)^j) = j$. Put

$$L_{p,n}(T) = \sum_{a=1}^{p-1}\sum_{j=0}^{p^{n-1}-1} \mu\big(\omega(a)(1+p)^j + p^n\mathbf{Z}_p\big)\,(1+T)^j = \sum_k a_{n,k}T^k. \tag{5}$$

Proposition 2.3. *Suppose $E[p]$ is irreducible, $n \ge 2$ and $1 \le k$. Then*

$$a_k \equiv a_{n,k} \pmod{p^{\,n-1-e_k}}, \qquad e_k = \max_{1\le i\le k} v_p(i).$$

In particular $a_k \equiv a_{n,k} \pmod{p^{n-1}}$ for $k < p$.

The proof is in Appendix A. The same bound, in a slightly different form, is [38, Prop. 3.5].[3]

For $n = 2$ the proposition gives every coefficient a_k with $k < p$ modulo p from finitely many rational numbers. Indeed $\omega(a) \equiv a^p \pmod{p^2}$, $\alpha \equiv a_p \pmod p$, and $[b/p]^+ = [a/p]^+$ when $b \equiv a \pmod p$, so, writing $b_{a,j}$ for the residue of $a^p(1+p)^j$ modulo p^2,

$$a_k \equiv \sum_{j=k}^{p-1}\binom{j}{k}\sum_{a=1}^{p-1}\Big(a_p^{-2}\Big[\frac{b_{a,j}}{p^2}\Big]^+ - a_p^{-3}\Big[\frac{a}{p}\Big]^+\Big) \pmod p \qquad (1 \le k < p). \tag{6}$$

At the irregular pairs the second level is not enough, and the same computation is carried out at level $n \ge 3$. The symmetry $[-r]^+ = [r]^+$ then halves the work.

Proposition 2.4. *Let $n \ge 2$, $1 \le k < p$, and let $b_{a,j}$ be the residue of $a^{p^{n-1}}(1+p)^j$ modulo p^n. Then*

$$a_k \equiv 2\sum_{j=k}^{p^{n-1}-1}\binom{j}{k}\sum_{a=1}^{(p-1)/2}\Big(\alpha^{-n}\Big[\frac{b_{a,j}}{p^n}\Big]^+ - \alpha^{-n-1}\Big[\frac{b_{a,j}}{p^{n-1}}\Big]^+\Big) \pmod{p^{n-1}}.$$

[3] Stein and Wuthrich compare consecutive Riemann sums through the divisibility of $L_{p,n+1} - L_{p,n}$ by $(1+T)^{p^{n-1}} - 1$; the proof in Appendix A instead estimates the oscillation of $(1+T)^{\ell(x)}$ on each residue class, which gives the bound in one step.

Proof. The Teichmüller character satisfies $\omega(a) \equiv a^{p^{n-1}} \pmod{p^n}$: indeed $a^{p^{n-1}} = \omega(a)^{p^{n-1}}\langle a\rangle^{p^{n-1}}$, and $\omega(a)^{p^{n-1}} = \omega(a)$ because $p^{n-1} \equiv 1 \pmod{p-1}$, while $\langle a\rangle^{p^{n-1}} \in 1 + p^n\mathbf{Z}_p$. So $b_{a,j}$ represents $\omega(a)(1+p)^j$ modulo p^n, and $\binom{j}{k} = 0$ for $j < k$. Since $\omega(p-a) = \omega(-1)\omega(a) = -\omega(a)$, we have $b_{p-a,j} \equiv -b_{a,j}$, and (3) together with Lemma 2.1(a) gives $\mu(-b + p^n\mathbf{Z}_p) = \mu(b + p^n\mathbf{Z}_p)$. The terms for a and $p-a$ in (5) are therefore equal, and Proposition 2.3 finishes the proof. □

For $n = 2$ this is (6), with $\alpha \equiv a_p \pmod{p}$. For $n = 3$ and $n = 4$ it requires α modulo p^2 and p^3, which Hensel's lemma gives from $\alpha \equiv a_p \pmod{p}$, and the symbols $[b/p^n]^+$ for about $\frac{1}{2}p^n$ residues b. At $p = 89$ and $n = 4$ this is $3.1 \cdot 10^7$ symbols, the largest computation in the paper.

The function $L_p(E,T)$ defined in (4) is the p-adic L-function of Mazur, Tate and Teitelbaum: it interpolates the twisted values $L(E,\chi,1)$. This is not used in the proofs, but it explains why Kato's theorem below, which concerns their function, applies to ours.

Proposition 2.5. *Let χ be a primitive even Dirichlet character of conductor p^n, $n \geq 1$, and $G(\chi) = \sum_{a \bmod p^n} \chi(a)e^{2\pi ia/p^n}$. Then*

$$\int_{\mathbf{Z}_p^\times} \chi \, d\mu = \frac{G(\chi)}{\alpha^n} \cdot \frac{L(E,\bar\chi,1)}{\Omega_E}.$$

In particular, if $n \geq 2$ and χ is a character of order p^{n-1} trivial on the roots of unity in $\mathbf{Z}_p^\times$, then $\zeta = \chi(1+p)$ is a primitive p^{n-1}-th root of unity and $L_p(E,\zeta-1)$ is this value.

Proof. Put $m = p^n$. The second term of (3) contributes $\alpha^{-n-1}\sum_{c \bmod p^{n-1}}[c/p^{n-1}]^+ \sum_{a\equiv c}\chi(a)$, and the inner sums vanish because χ is primitive of conductor m. So $\int \chi\, d\mu = \alpha^{-n}\sum_a \chi(a)[a/m]^+$. Next, $\lambda(a/m) = 2\pi i\int_0^{i\infty} f(\tau + a/m)\,d\tau$, and the orthogonality of additive characters gives

$$\sum_{a \bmod m} \chi(a) f\Big(\tau + \frac{a}{m}\Big) = \sum_{n'\geq 1} a_{n'}q^{n'} \sum_a \chi(a)e^{2\pi i a n'/m} = G(\chi)\sum_{n'} \bar\chi(n')a_{n'}q^{n'},$$

the last step because $\sum_a \chi(a)e^{2\pi i a n'/m} = \bar\chi(n')G(\chi)$ for primitive χ. Integrating, $\sum_a \chi(a)\lambda(a/m) = -G(\chi)L(E,\bar\chi,1)$, with the normalisation of the footnote above. As χ is even, $\sum_a \chi(a)\lambda(-a/m) = \sum_a \chi(a)\lambda(a/m)$, so $\sum_a \chi(a)\lambda^+(a/m) = G(\chi)L(E,\bar\chi,1)$. Dividing by Ω_E proves the formula. For the last statement, such a character is even, it has conductor p^n, and $\chi(x) = \chi(\langle x\rangle) = \zeta^{\ell(x)}$, and $\int \zeta^{\ell(x)}d\mu = L_p(E,\zeta-1)$ by (4). □

The two formulas (6) and Proposition 2.4 are what the proofs of Theorems A–D evaluate.

3. p-adic heights

Let p be good ordinary for E. The cyclotomic p-adic height was defined by Schneider [32] and Perrin-Riou; we use the explicit description of Mazur and Tate [26] in the normalisation of [38, §4], and compute it by the algorithm of Mazur, Stein and Tate [25]. Let $t = -x/y$ be the parameter at the origin of a minimal model, $\omega_E = dx/(2y + a_1x + a_3)$ and $b_2 = a_1^2 + 4a_2$. The *canonical p-adic sigma function* $\sigma_p(t) \in t + t^2\mathbf{Z}_p[[t]]$ is the unique function of the form $\sigma_p = t + \cdots$, odd for the inverse map of the formal group, satisfying

$$x + c = -\frac{d}{\omega_E}\Big(\frac{1}{\sigma_p}\frac{d\sigma_p}{\omega_E}\Big), \qquad c = \frac{b_2 - e_2}{12},$$

where $e_2 = E_2(E,\omega_E)$ is the value of Katz's p-adic Eisenstein series of weight two [17]; Mazur and Tate proved that σ_p has coefficients in $\mathbf{Z}_p$ [26]. Appendix D gives σ_p in closed form,

characterises c through Frobenius, and works the example of E_4 at 5. If $P \in E(\mathbf{Q})$ reduces to the identity modulo p and to the identity component of the Néron model at every prime, and $e(P)$ is the positive square root of the denominator of $x(P)$, put

$$\hat{h}_p(P) = 2\log_p \frac{e(P)}{\sigma_p(P)},$$

and extend to all of $E(\mathbf{Q})$ by $\hat{h}_p(mP) = m^2\hat{h}_p(P)$. This is a quadratic form, and

$$\langle P, Q\rangle_p = \tfrac{1}{2}\big(\hat{h}_p(P+Q) - \hat{h}_p(P) - \hat{h}_p(Q)\big)$$

is a symmetric bilinear form on $E(\mathbf{Q})/E(\mathbf{Q})_{\mathrm{tors}}$ with values in $\mathbf{Q}_p$. For points $P_1, \dots, P_r$ put

$$\mathrm{Reg}_p(P_1, \dots, P_r) = \det\big(\langle P_i, P_j\rangle_p\big), \qquad \mathrm{Reg}_\gamma = \frac{\mathrm{Reg}_p}{\log_p(1+p)^r},$$

and write $\mathrm{Reg}_p(E/\mathbf{Q})$, $\mathrm{Reg}_\gamma(E/\mathbf{Q})$ for the values on a basis of $E(\mathbf{Q})/E(\mathbf{Q})_{\mathrm{tors}}$; they are well defined up to sign. The division by $\log_p(1+p)^r$ matches the choice of the generator $1+p$ of $1+p\mathbf{Z}_p$ that identifies the Iwasawa algebra with Λ [38, §4.4].

Lemma 3.1. *If $P_1, \dots, P_r$ are independent and generate a subgroup of index n in $E(\mathbf{Q})/E(\mathbf{Q})_{\mathrm{tors}}$, a free group of rank r, then $\mathrm{Reg}_p(P_1, \dots, P_r) = n^2\,\mathrm{Reg}_p(E/\mathbf{Q})$.*

Proof. If $(P_i) = A\,(Q_j)$ with (Q_j) a basis and A an integral matrix, then $|\det A| = n$ and the Gram matrices satisfy $G_P = AG_QA^{\mathsf{T}}$. □

The size of the regulator is controlled by the reduction of E at p and at the bad primes, and the following lemma explains the valuations that appear in the examples below.

Lemma 3.2. *Let $m = \mathrm{lcm}\big(\#\tilde{E}(\mathbf{F}_p), c_v : v \mid N\big)$ and $k = v_p(m)$. Then $\langle P, Q\rangle_p \in p^{1-2k}\mathbf{Z}_p$ for all $P, Q \in E(\mathbf{Q})$, and $v_p(\mathrm{Reg}_p(P_1, \dots, P_r)) \geq r(1-2k)$. In particular, if p is not anomalous and divides no Tamagawa number, then $\mathrm{Reg}_\gamma(P_1, \dots, P_r) \in \mathbf{Z}_p$.*

Proof. Let $R = mP$, which reduces to the identity modulo p and to the identity component at every bad prime. Write $x(R) = a/e^2$ and $y(R) = b/e^3$ in lowest terms, with $e = e(R)$. If $v_p(x(R)) = -2j$ with $j \geq 1$, then $v_p(e) = j$ and $p \nmid ab$, and $t(R) = -x/y = -ae/b$ has valuation j. So

$$\frac{e(R)}{\sigma_p(R)} = \frac{e}{t(R)} \cdot \frac{t(R)}{\sigma_p(R)} = -\frac{b}{a} \cdot \Big(\frac{\sigma_p(t)}{t}\Big)^{-1}\Big|_{t=t(R)},$$

and both factors are units of $\mathbf{Z}_p$: the first because $p \nmid ab$, the second because $\sigma_p(t)/t \in 1+t\mathbf{Z}_p[[t]]$ and $t(R) \in p\mathbf{Z}_p$. For any $u \in \mathbf{Z}_p^\times$ we have $u^{p-1} \in 1+p\mathbf{Z}_p$, so $\log_p u = \frac{1}{p-1}\log_p(u^{p-1}) \in p\mathbf{Z}_p$. Hence $\hat{h}_p(R) \in p\mathbf{Z}_p$, and $\hat{h}_p(P) = m^{-2}\hat{h}_p(R) \in p^{1-2k}\mathbf{Z}_p$. The pairing is half a sum of three such values, and p is odd. The bound on the determinant follows by expanding it, and the last statement from $k = 0$ and $v_p(\log_p(1+p)) = 1$. □

So when p is not anomalous and divides no Tamagawa number, the factors $(1-\alpha^{-1})^2$ and $\prod c_v$ are units, and the pair (E, p) is irregular exactly when p^{r+1} divides $\mathrm{Reg}_p(E/\mathbf{Q})$, that is, when the matrix $p^{-1}(\langle P_i, P_j\rangle_p)$ is singular modulo p. If this matrix behaved like a random symmetric matrix over $\mathbf{F}_p$, it would be singular with probability close to $1/p$. This is the heuristic behind §9, and Figure 7 bears it out. When p is anomalous, $k \geq 1$, and the entries may have negative valuation, as they do for E_4 at 5 below.

Schneider's conjecture asserts that $\mathrm{Reg}_p(E/\mathbf{Q}) \neq 0$ for every p. For curves of rank one with complex multiplication it is a theorem of Bertrand, proved by transcendence methods; beyond that case no general result is known.[4]

3.1. **Computing the pairing.** Three steps turn the definition into numbers [25]. The value e_2 is read off from the eigenline of Frobenius with unit eigenvalue in the de Rham cohomology $H^1_{\mathrm{dR}}(E/\mathbf{Q}_p)$, written in the basis $\omega,\ x\omega$; the matrix of Frobenius on this two-dimensional space is computed by Kedlaya's algorithm, and that eigenline is found by iterating it. The function σ_p is then given by the closed formula of Proposition D.2, to any prescribed t-adic and p-adic precision. Finally, for a point P one takes an integer m such that mP reduces to the identity modulo p and to the identity component at the primes of bad reduction; $m = \mathrm{lcm}(\#\tilde{E}(\mathbf{F}_p), c_v)$ will do, since the component group at v has order c_v. Then $\hat{h}_p(P) = m^{-2}\hat{h}_p(mP)$, and mP lies in the domain of convergence of σ_p because $t(mP) \in p\mathbf{Z}_p$.

For E_4 and $p = 5$ one may take $m = 10$, as $\#\tilde{E}_4(\mathbf{F}_5) = 10$ and $c_2 = 2$. The resulting Gram matrix has entries of valuation -1, 0 and 1; for instance

$$\langle P_1, P_1\rangle_5 = 2\cdot 5^{-1} + 1 + 5 + 2\cdot 5^2 + O(5^3), \qquad \langle P_1, P_2\rangle_5 = 4 + 4\cdot 5 + 4\cdot 5^2 + O(5^3).$$

Clearing the denominator, the integer matrix

$$N = \begin{pmatrix} 51532 & 28745 & 18034 & 44467 \\ 28745 & 25950 & 46565 & 66895 \\ 18034 & 46565 & 64483 & 37404 \\ 44467 & 66895 & 37404 & 20752 \end{pmatrix}$$

satisfies $N \equiv 5\big(\langle P_i, P_j\rangle_5\big)_{i,j} \pmod{5^7}$. Its determinant is $-1\,409\,401\,157\,357\,234\,375$, which is $2\cdot 5^6$ modulo 5^7, so

$$\mathrm{Reg}_5(P_1,\dots,P_4) = 5^{-4}\det N + O(5^3) = 2\cdot 5^2 + O(5^3).$$

Since every entry has valuation at least -1, the determinant could have valuation as low as -4; it has valuation 2, so six powers of 5 cancel in its expansion. In Section 8.2 the valuation 2 of Reg_5, the valuation -4 of $\log_5(6)^{-4}$ and the valuation 2 of the anomalous factor $(1-\alpha^{-1})^2$ add up to 0, and this is what makes $v_5(S_5) = 0$.

4. Iwasawa theory

Let $\mathbf{Q}_\infty$ be the unique $\mathbf{Z}_p$-extension of $\mathbf{Q}$ contained in $\mathbf{Q}(\mu_{p^\infty})$, with layers $\mathbf{Q}_n$ of degree p^n and Galois group Γ (Figure 3). The element $\gamma \in \Gamma$ acting on μ_{p^∞} through the cyclotomic character by $1+p$ is a topological generator, and $\gamma \mapsto 1+T$ identifies $\mathbf{Z}_p[[\Gamma]]$ with Λ. Let

$$\mathrm{Sel}_{p^\infty}(E/\mathbf{Q}_\infty) = \varinjlim_n \mathrm{Sel}_{p^\infty}(E/\mathbf{Q}_n), \qquad X(E/\mathbf{Q}_\infty) = \mathrm{Hom}\big(\mathrm{Sel}_{p^\infty}(E/\mathbf{Q}_\infty), \mathbf{Q}_p/\mathbf{Z}_p\big),$$

a finitely generated Λ-module. When p is good ordinary, Kato proved that $X(E/\mathbf{Q}_\infty)$ is Λ-torsion [16, Thm. 17.4], using Rohrlich's nonvanishing theorem [31]; let $f_E(T) \in \Lambda$ be a generator of its characteristic ideal, well defined up to a unit of Λ.

Theorem 4.1 (Kato)**.** *Let $p \geq 5$ be good ordinary for E and suppose that $\bar{\rho}_{E,p}$ is surjective. Then $f_E(T)$ divides $L_p(E,T)$ in Λ.*

[4] See [38] for references. For a given curve and prime the nonvanishing is decided by computing the pairing to sufficient precision, which is how it is established for the curves of this paper.

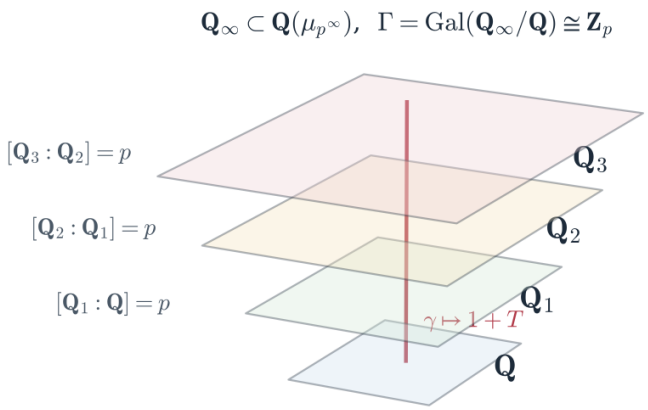


FIGURE 3. The cyclotomic $\mathbf{Z}_p$-extension. The Selmer groups of E over the layers $\mathbf{Q}_n$ are assembled into a module over $\Lambda = \mathbf{Z}_p[[\Gamma]]$, with a generator γ of Γ acting as $1+T$.

This is [16, Thm. 17.4], in the form stated in [38, Thm. 7.3].[5] Under the hypotheses of [37] the divisibility is an equality, but this is not used.

Theorem 4.2 (Schneider, Perrin-Riou)**.** *Let p be good ordinary for E and $r = \operatorname{rank} E(\mathbf{Q})$. Then $\operatorname{ord}_{T=0} f_E(T) \geq r$, with equality if and only if $\operatorname{Reg}_p(E/\mathbf{Q}) \neq 0$ and $Ш(E/\mathbf{Q})[p^\infty]$ is finite. If equality holds, the coefficient f_r of T^r in f_E satisfies*

$$v_p(f_r) = v_p\Big(\big(1-\alpha^{-1}\big)^2 \cdot \frac{\prod_v c_v \cdot \#Ш(E/\mathbf{Q})[p^\infty] \cdot \operatorname{Reg}_\gamma(E/\mathbf{Q})}{\#E(\mathbf{Q})[p^\infty]^2}\Big).$$

This is due to Schneider [33] and Perrin-Riou [30]; see [38, Thm. 6.1].[6]

4.1. **Where the inequality comes from.** The inequality in Theorem 4.2 has a short proof from the structure of Λ-modules, and it shows where each factor of the leading term comes from.

For a number field F the Selmer group $\mathrm{Sel}_{p^\infty}(E/F) \subset H^1(F, E[p^\infty])$ consists of the classes whose restriction to every completion lies in the image of the local Kummer map, and it sits

[5] Kato's theorem is stated for the p-adic representation $\rho_{E,p}$ on the Tate module. For $p \geq 5$ surjectivity of $\bar\rho_{E,p}$ implies surjectivity of $\rho_{E,p}$: a closed subgroup of $\mathrm{GL}_2(\mathbf{Z}_p)$ whose reduction is $\mathrm{GL}_2(\mathbf{F}_p)$ contains $\mathrm{SL}_2(\mathbf{Z}_p)$ [34, IV-23, Lemma 3], and the determinant is the cyclotomic character, which is onto $\mathbf{Z}_p^\times$.

[6] The inequality comes from Mazur's control theorem [21]: the Γ-coinvariants of $X(E/\mathbf{Q}_\infty)$ have $\mathbf{Z}_p$-rank at least the corank of $\mathrm{Sel}_{p^\infty}(E/\mathbf{Q})$, which is at least r. The factor $(1-\alpha^{-1})^2$ has valuation $2v_p(\#\tilde E(\mathbf{F}_p))$, so it is a unit unless p is anomalous, $a_p \equiv 1 \pmod p$.

in the exact sequence

$$0 \to E(F) \otimes \mathbf{Q}_p/\mathbf{Z}_p \to \mathrm{Sel}_{p^\infty}(E/F) \to Ш(E/F)[p^\infty] \to 0.$$

Hence its $\mathbf{Z}_p$-corank s_F satisfies $s_F \geq \operatorname{rank} E(F)$, with equality if and only if $Ш(E/F)[p^\infty]$ is finite. Mazur's control theorem [21] (see also [14]) states that for p good ordinary the restriction map

$$\mathrm{Sel}_{p^\infty}(E/\mathbf{Q}) \longrightarrow \mathrm{Sel}_{p^\infty}(E/\mathbf{Q}_\infty)^\Gamma$$

has finite kernel and cokernel. Dualising, $X(E/\mathbf{Q}_\infty)_\Gamma = X/TX$ has $\mathbf{Z}_p$-rank $s_{\mathbf{Q}} \geq r$.

On the other side, by the structure theorem there is a homomorphism $X \to \bigoplus_i \Lambda/(g_i)$ with finite kernel and cokernel, where $g_i \in \Lambda$ and $f_E = \prod_i g_i$ up to a unit. Finite modules do not change $\mathbf{Z}_p$-ranks of T-coinvariants, and $\Lambda/(g_i, T) = \mathbf{Z}_p/(g_i(0))$ has rank one if $T \mid g_i$ and rank zero otherwise. Therefore

$$r \leq s_{\mathbf{Q}} = \operatorname{rank}_{\mathbf{Z}_p} X/TX = \#\{i : \ T \mid g_i\} \leq \operatorname{ord}_T f_E.$$

Equality throughout requires two things: $s_{\mathbf{Q}} = r$, that is, $Ш(E/\mathbf{Q})[p^\infty]$ finite; and that T divide each g_i at most once, so that T acts semisimply on the T-primary part of X. Schneider showed that the second condition is governed by the p-adic height: when $Ш(E/\mathbf{Q})[p^\infty]$ is finite, the map from the Γ-invariants to the Γ-coinvariants of the Selmer group is given on $E(\mathbf{Q}) \otimes \mathbf{Z}_p$ by the height pairing, so semisimplicity is equivalent to $\mathrm{Reg}_p(E/\mathbf{Q}) \neq 0$ [33]. When both conditions hold, the leading coefficient f_r is computed by following orders of finite groups through the control theorem. The cokernel of restriction contributes the Tamagawa numbers at the primes of bad reduction and the factor $\#\tilde{E}(\mathbf{F}_p)[p^\infty]^2$ at p, which for an ordinary prime equals the p-part of $(1-\alpha^{-1})^2$ up to a unit [14]; the finite group $Ш(E/\mathbf{Q})[p^\infty]$ contributes its order, and the height pairing contributes Reg_γ.

5. The criterion

For independent points $P_1, \ldots, P_r \in E(\mathbf{Q})$ and a good ordinary prime p put

$$S_p(P_1, \ldots, P_r) = \big(1 - \alpha^{-1}\big)^2 \cdot \frac{\prod_v c_v \cdot \mathrm{Reg}_\gamma(P_1, \ldots, P_r)}{\#E(\mathbf{Q})^2_{\mathrm{tors}}} \in \mathbf{Q}_p. \tag{7}$$

The points are *p-saturated* if no point $Q \in E(\mathbf{Q})$ satisfies $pQ \in \sum \mathbf{Z}P_i + E(\mathbf{Q})_{\mathrm{tors}}$ unless $Q \in \sum \mathbf{Z}P_i + E(\mathbf{Q})_{\mathrm{tors}}$.

Theorem 5.1. *Let $E/\mathbf{Q}$ be an elliptic curve and $p \geq 5$ a good ordinary prime with $\bar{\rho}_{E,p}$ surjective. Let $P_1, \ldots, P_r \in E(\mathbf{Q})$ be independent and p-saturated, and let a_r be the coefficient of T^r in $L_p(E,T)$. If*

$$v_p(a_r) = v_p\big(S_p(P_1, \ldots, P_r)\big) < \infty, \tag{8}$$

then:

(i) $\operatorname{rank} E(\mathbf{Q}) = r$, *and $P_1, \ldots, P_r$ generate a subgroup of $E(\mathbf{Q})/E(\mathbf{Q})_{\mathrm{tors}}$ of index prime to p;*
(ii) $Ш(E/\mathbf{Q})[p^\infty] = 0$;
(iii) $\mathrm{Reg}_p(E/\mathbf{Q}) \neq 0$ *and* $\operatorname{ord}_{T=0} L_p(E,T) = r$;
(iv) $L_p(E,T) = f_E(T)\, u(T)$ *with $u \in \Lambda^\times$; that is, the main conjecture of Iwasawa theory holds for (E,p).*

Proof. Surjectivity of $\bar{\rho}_{E,p}$ gives three facts used repeatedly: $E[p]$ is irreducible, so $L_p(E,T) \in \Lambda$ by §2; $E(\mathbf{Q})[p] = 0$, so $\#E(\mathbf{Q})[p^\infty] = 1$ and $\#E(\mathbf{Q})_{\mathrm{tors}}$ is prime to p; and Theorem 4.1 applies. Write $L_p = f_E\, d$ with $d \in \Lambda$, and let $\rho = \operatorname{rank} E(\mathbf{Q}) \geq r$, the inequality because the P_i are independent.

By Theorem 4.2, $\operatorname{ord}_T f_E \geq \rho \geq r$, so the coefficients of $T^0, \dots, T^{r-1}$ of $L_p = f_E d$ vanish and the coefficient of T^r is

$$a_r = f_r\, d(0),$$

where f_r is the coefficient of T^r in f_E (zero if $\operatorname{ord}_T f_E > r$). By (8), $a_r \neq 0$. Hence $f_r \neq 0$, so $\operatorname{ord}_T f_E = r$; together with $\operatorname{ord}_T f_E \geq \rho \geq r$ this gives $\rho = r$ and, by the equality case of Theorem 4.2, $\operatorname{Reg}_p(E/\mathbf{Q}) \neq 0$ and $Ш(E/\mathbf{Q})[p^\infty]$ finite. Also $d(0) \neq 0$ and $\operatorname{ord}_T L_p = r$. This proves (iii) and the first half of (i).

Since $\rho = r$, the points P_i generate a subgroup of finite index n in $E(\mathbf{Q})/E(\mathbf{Q})_{\mathrm{tors}}$. Suppose $p \mid n$. The quotient then has an element of order p, represented by some $Q \notin \sum \mathbf{Z}P_i + E(\mathbf{Q})_{\mathrm{tors}}$ with $pQ \in \sum \mathbf{Z}P_i + E(\mathbf{Q})_{\mathrm{tors}}$, contradicting p-saturation. So $p \nmid n$, completing (i), and by Lemma 3.1, $v_p(S_p(P_1, \dots, P_r)) = v_p(S_p(E))$, where $S_p(E)$ is (7) evaluated on a basis.

Theorem 4.2 now gives $v_p(f_r) = v_p(S_p(E)) + v_p\#Ш(E/\mathbf{Q})[p^\infty]$. Since $d(0) \in \mathbf{Z}_p$,

$$v_p(S_p(E)) + v_p\#Ш(E/\mathbf{Q})[p^\infty] = v_p(f_r) = v_p(a_r) - v_p(d(0)) \leq v_p(a_r) = v_p(S_p(E)),$$

so $\#Ш(E/\mathbf{Q})[p^\infty] = 1$ and $v_p(d(0)) = 0$. This is (ii), and $d(0) \in \mathbf{Z}_p^\times$ means $d \in \Lambda^\times$, which is (iv). □

Remark 5.2. The proof does not use the coefficients $a_0, \dots, a_{r-1}$: their vanishing is a consequence of Theorems 4.1 and 4.2. When they are computed, their vanishing modulo the available power of p is a check on the computation, and it held in every case treated below. Only a_r modulo $p^{v_p(S)+1}$ is needed, and Proposition 2.3 with $n = v_p(S) + 2$ supplies it when $r < p$. A pair with $v_p(S) > 0$ is called *irregular*; for those a deeper Riemann sum is required.

Remark 5.3. The quantity $S_p(E)$ is a product of factors that need not be integral individually, but the product is constrained. Suppose $\operatorname{ord}_T f_E = \operatorname{rank} E(\mathbf{Q})$. Since $f_E \in \Lambda$, its leading coefficient lies in $\mathbf{Z}_p$, and Theorem 4.2 gives

$$v_p\big(S_p(E)\big) + v_p\#Ш(E/\mathbf{Q})[p^\infty] \geq 0.$$

So a negative $v_p(S_p(E))$ would force $Ш(E/\mathbf{Q})[p] \neq 0$, and whenever Theorem 5.1 applies, $v_p(S_p(E)) = v_p(a_r) \geq 0$. At an anomalous prime the Euler factor $(1 - \alpha^{-1})^2$ has valuation at least 2, and the inequality allows $\operatorname{Reg}_\gamma$ to have a pole of the same order; this is what happens for E_4 at $p = 5$ in §8.2.

Proposition 5.4. *If the conclusions (i) and (ii) of Theorem 5.1 hold for E and p, and E' is isogenous to E, then* $\operatorname{rank} E'(\mathbf{Q}) = r$ *and* $Ш(E'/\mathbf{Q})[p^\infty] = 0$.

Proof. Isogenous curves have the same rank. By the first paragraph of the proof of Lemma 2.1(c) there is an isogeny $\psi\colon E \to E'$ of degree d prime to p; it induces $\psi_*\colon Ш(E/\mathbf{Q})[p^\infty] \to Ш(E'/\mathbf{Q})[p^\infty]$, and $\hat{\psi}_*\psi_*$, $\psi_*\hat{\psi}_*$ are multiplication by d, which is invertible on p-primary groups. So ψ_* is an isomorphism. □

6. Surjectivity from Frobenius traces

Proposition 6.1. *Let $p \geq 5$ and suppose there are primes $q_1, q_2, q_3 \neq p$ of good reduction for E such that $a_{q_i} \not\equiv 0 \pmod p$ and*

(1) *$a_{q_1}^2 - 4q_1$ is a nonsquare modulo p;*
(2) *$a_{q_2}^2 - 4q_2$ is a nonzero square modulo p;*
(3) *$u = a_{q_3}^2/q_3 \bmod p$ satisfies $u \notin \{0, 1, 2, 4\}$ and $u^2 - 3u + 1 \not\equiv 0 \pmod p$.*

Then $\bar{\rho}_{E,p}$ is surjective.

Proof. Let $G \subseteq \mathrm{GL}_2(\mathbf{F}_p)$ be the image. Since $\det \bar\rho_{E,p}$ is the mod-p cyclotomic character, $\det(G) = \mathbf{F}_p^\times$, so $G = \mathrm{GL}_2(\mathbf{F}_p)$ if and only if $G \supseteq \mathrm{SL}_2(\mathbf{F}_p)$. By Proposition C.1 in Appendix C, if $G \not\supseteq \mathrm{SL}_2(\mathbf{F}_p)$ then G is contained in a Borel subgroup, in the normaliser of a split Cartan subgroup, or in the normaliser of a nonsplit Cartan subgroup, or every element of its image in $\mathrm{PGL}_2(\mathbf{F}_p)$ has order at most 5. The element $g_i = \bar\rho_{E,p}(\mathrm{Frob}_{q_i})$ has characteristic polynomial $X^2 - a_{q_i}X + q_i$ modulo p, and trace $a_{q_i} \neq 0$.

Borel and split Cartan. The characteristic polynomial of g_1 is irreducible over $\mathbf{F}_p$, so g_1 has no eigenvector and lies in no Borel subgroup; a split Cartan subgroup is contained in a Borel subgroup; and the elements of the normaliser of a split Cartan subgroup outside it have trace zero. So g_1 is in neither.

Nonsplit Cartan. A nonsplit Cartan subgroup is the multiplicative group of a field $\mathbf{F}_{p^2} \subset M_2(\mathbf{F}_p)$; its elements are scalar or have irreducible characteristic polynomial. The element g_2 has two distinct eigenvalues in $\mathbf{F}_p$, so it is not in the Cartan subgroup, and its trace is nonzero, so it is not in the other coset of the normaliser.

Small projective image. By Lemma 6.2 the image of g_3 in $\mathrm{PGL}_2(\mathbf{F}_p)$ has order at least 6. □

Lemma 6.2. *Let $g \in \mathrm{GL}_2(\mathbf{F}_p)$, $p \geq 5$, with $\operatorname{tr} g \neq 0$, and put $u = (\operatorname{tr} g)^2/\det g$. If the image of g in $\mathrm{PGL}_2(\mathbf{F}_p)$ has order $m \leq 5$, then $u = 4, 0, 1, 2$ for $m = 1, 2, 3, 4$ respectively, and $u^2 - 3u + 1 = 0$ for $m = 5$.*

Proof. If g is not semisimple it has a single eigenvalue λ and $u = 4\lambda^2/\lambda^2 = 4$. Otherwise let λ, λ' be its eigenvalues in $\mathbf{F}_{p^2}$ and $\zeta = \lambda/\lambda'$. Then $u = (\lambda + \lambda')^2/(\lambda\lambda') = \zeta + 2 + \zeta^{-1}$, and the image of g has order equal to the order of ζ. For ζ of order $1, 2, 3, 4$ one finds $u = 4, 0, 1, 2$. For ζ of order 5, $w = \zeta + \zeta^{-1}$ satisfies $w^2 + w - 1 = 0$, and with $u = w + 2$ this reads $u^2 - 3u + 1 = 0$. □

Proposition 6.1 reduces surjectivity to three point counts. It also excludes complex multiplication, since for $p \geq 5$ the image of $\bar\rho_{E,p}$ for a curve with CM lies in the normaliser of a Cartan subgroup. The triple (q_1, q_2, q_3) is called a *surjectivity witness*. The search for one always ends when $\bar\rho_{E,p}$ is surjective, and never ends otherwise.

Proposition 6.3. *Let $p \geq 5$. Then $\bar\rho_{E,p}$ is surjective if and only if a surjectivity witness exists, and in that case each of the conditions* (1), (2), (3) *of Proposition 6.1 holds for a set of primes q of positive density.*

Proof. One direction is Proposition 6.1. Suppose $\bar\rho_{E,p}$ is surjective, let $K = \mathbf{Q}(E[p])$, so that $\mathrm{Gal}(K/\mathbf{Q}) \cong \mathrm{GL}_2(\mathbf{F}_p)$, and let $q \neq p$ be a prime of good reduction, unramified in K. The conditions on q depend only on the conjugacy class of Frob_q in $\mathrm{GL}_2(\mathbf{F}_p)$, through its trace a_q and determinant q. By the Chebotarev density theorem [28, Ch. VII, Thm. 13.4] every conjugacy class C is the Frobenius class of a set of primes of density $\#C/\#\mathrm{GL}_2(\mathbf{F}_p) > 0$, so it suffices to exhibit one element of $\mathrm{GL}_2(\mathbf{F}_p)$ for each condition.

For (1), take g to be multiplication by an element $\lambda \in \mathbf{F}_{p^2} \setminus \mathbf{F}_p$ with $\lambda + \lambda^p \neq 0$, acting on $\mathbf{F}_{p^2} \cong \mathbf{F}_p^2$. Such λ exist, since the λ with $\lambda^{p-1} = -1$ number $p - 1 < p^2 - p$. The characteristic polynomial of g is that of λ over $\mathbf{F}_p$, which is irreducible, so its discriminant is a nonsquare, and its trace $\lambda + \lambda^p$ is nonzero.

For (2), take $g = \mathrm{diag}(1, 2)$: trace $3 \neq 0$ and discriminant 1.

For (3), Lemma 6.2 and its proof show that for semisimple g with eigenvalue ratio ζ the value $u = \zeta + 2 + \zeta^{-1}$ determines $\zeta + \zeta^{-1}$, hence the pair $\{\zeta, \zeta^{-1}\}$, hence the order of ζ; so u avoids $0, 1, 2, 4$ and the roots of $u^2 - 3u + 1$ exactly when ζ has order at least 6, and then $u \neq 0$ means $\operatorname{tr} g \neq 0$. If $p \geq 7$, take $g = \mathrm{diag}(1, x)$ with x a generator of $\mathbf{F}_p^\times$, of order $p - 1 \geq 6$. If $p = 5$, take g to be multiplication by $\lambda \in \mathbf{F}_{25}^\times$ with λ^4 of order 6; this is possible because

$\lambda \mapsto \lambda^{p-1}$ maps $\mathbf{F}_{p^2}^{\times}$ onto the elements of norm one, a cyclic group of order $p+1=6$. The eigenvalues of g are λ, λ^5 with ratio λ^{-4}, of order 6. □

Remark 6.4. In the computations the witnesses were small: every rank-three pair in Theorem B has a witness with all $q_i \le 107$, the largest being needed for the class $92651a$ at $p=5$; those for E_4, listed in Table 2, have $q_i \le 67$, and those for E_5 and E_6 have $q_i \le 19$.

The same idea gives irreducibility of $E[\ell]$ for a single prime q: if $X^2 - a_q X + q$ is irreducible modulo ℓ (for $\ell = 2$: if a_q and q are odd), then $\bar\rho_{E,p}$ at ℓ fixes no line. Mazur's theorem [22] says that an elliptic curve over $\mathbf{Q}$ with a rational isogeny of prime degree ℓ has $\ell \in \{2,3,5,7,11,13,17,19,37,43,67,163\}$, so a witness for each of these twelve primes shows that the isogeny class of E consists of E alone.

7. Saturation from reductions

Proposition 7.1. *Let $P_1,\dots,P_r \in E(\mathbf{Q})$, let p be a prime with $E(\mathbf{Q})[p]=0$, and let $\mathcal{L}$ be a set of primes $\ell \neq p$ of good reduction such that for each $\ell \in \mathcal{L}$ the p-Sylow subgroup of $\tilde E(\mathbf{F}_\ell)$ is cyclic and nontrivial. Fix an isomorphism $\phi_\ell\colon \tilde E(\mathbf{F}_\ell)/p\tilde E(\mathbf{F}_\ell) \to \mathbf{F}_p$. If the vectors*

$$\big(\phi_\ell(\tilde P_1),\dots,\phi_\ell(\tilde P_r)\big) \in \mathbf{F}_p^r, \qquad \ell \in \mathcal{L},$$

span $\mathbf{F}_p^r$, then $P_1,\dots,P_r$ are p-saturated.

Proof. Suppose $pQ = \sum n_i P_i + T$ with T torsion. As $E(\mathbf{Q})_{\mathrm{tors}}$ has order prime to p, $T = pT'$ for a torsion point T', and $p(Q-T') = \sum n_i P_i$. Reduction modulo ℓ is a homomorphism $E(\mathbf{Q}) \to \tilde E(\mathbf{F}_\ell)$, so $\sum n_i \tilde P_i \in p\tilde E(\mathbf{F}_\ell)$ and $\sum n_i \phi_\ell(\tilde P_i) = 0$ for every $\ell \in \mathcal{L}$. The spanning condition forces $n_i \equiv 0 \pmod p$, say $n_i = pm_i$; then $p(Q - T' - \sum m_i P_i) = 0$, so $Q - T' - \sum m_i P_i$ is torsion and $Q \in \sum \mathbf{Z}P_i + E(\mathbf{Q})_{\mathrm{tors}}$. □

The map ϕ_ℓ is explicit. If $\#\tilde E(\mathbf{F}_\ell) = p^k m$ with $p \nmid m$ and the p-Sylow subgroup is cyclic, then a point R lies in $p\tilde E(\mathbf{F}_\ell)$ if and only if $(\#\tilde E(\mathbf{F}_\ell)/p)R = 0$; so $\phi_\ell(R)$ is the discrete logarithm of $(\#\tilde E(\mathbf{F}_\ell)/p)R$ in the subgroup of order p. Appendix E carries this out for E_4 and $p=5$.

Like the search for a surjectivity witness, the search for a set $\mathcal{L}$ is a decision procedure: it succeeds whenever the points are saturated. The argument is Kummer theory over the field $K = \mathbf{Q}(E[p])$, and it rests on a vanishing result for group cohomology.

Lemma 7.2. *Let G be a finite group, M a G-module, and z a central element of G such that $z-1$ is invertible on M. Then $H^1(G,M)=0$.*

Proof. Let f be a 1-cocycle. Since $zg = gz$, the cocycle relation gives $f(z) + zf(g) = f(g) + gf(z)$, that is, $(z-1)f(g) = (g-1)f(z)$. Hence $f(g) = (g-1)m$ with $m = (z-1)^{-1}f(z)$, and f is a coboundary. □

Proposition 7.3. *Let $p \ge 5$ with $\bar\rho_{E,p}$ surjective, and let $P_1,\dots,P_r \in E(\mathbf{Q})$ be p-saturated. Then there are infinitely many sets $\mathcal{L}$ of r primes satisfying the hypotheses of Proposition 7.1.*

Proof. Put $G = \mathrm{Gal}(K/\mathbf{Q}) \cong \mathrm{GL}_2(\mathbf{F}_p)$. The scalar matrix 2 is central and $2-1=1$ is invertible, so $H^1(G, E[p]) = 0$ by Lemma 7.2. In the inflation–restriction sequence

$$0 \to H^1(G,E[p]) \to H^1(\mathbf{Q},E[p]) \to H^1(K,E[p])^G$$

the restriction map is therefore injective. Composing with the Kummer map $E(\mathbf{Q})/pE(\mathbf{Q}) \hookrightarrow H^1(\mathbf{Q},E[p])$ gives an injection

$$\kappa\colon E(\mathbf{Q})/pE(\mathbf{Q}) \hookrightarrow H^1(K,E[p])^G = \mathrm{Hom}_G\big(\mathrm{Gal}(\bar{\mathbf{Q}}/K), E[p]\big), \qquad \kappa(P)(\sigma) = \sigma(Q) - Q,$$

where $pQ = P$. Saturation, together with $E(\mathbf{Q})[p] = 0$ and $p \nmid \#E(\mathbf{Q})_{\mathrm{tors}}$, says that $P_1, \dots, P_r$ are linearly independent in $E(\mathbf{Q})/pE(\mathbf{Q})$.

Let $L = K(Q_1, \dots, Q_r)$ with $pQ_i = P_i$, and consider $\Phi = (\kappa(P_1), \dots, \kappa(P_r))\colon \mathrm{Gal}(L/K) \hookrightarrow E[p]^r$. Its image H is a G-submodule. Since $E[p]$ is absolutely irreducible with $\mathrm{End}_G(E[p]) = \mathbf{F}_p$, every G-submodule of $E[p]^r = E[p] \otimes_{\mathbf{F}_p} \mathbf{F}_p^r$ has the form $E[p] \otimes W$ for a subspace $W \subseteq \mathbf{F}_p^r$. If $W \neq \mathbf{F}_p^r$, choose $(c_1, \dots, c_r) \neq 0$ orthogonal to W; then $\kappa(\sum c_i P_i) = \sum c_i \kappa(P_i)$ vanishes on $\mathrm{Gal}(L/K)$, hence on $\mathrm{Gal}(\bar{\mathbf{Q}}/K)$, so $\sum c_i P_i \in pE(\mathbf{Q})$, which contradicts independence. So Φ is an isomorphism $\mathrm{Gal}(L/K) \cong E[p]^r$.

Fix $g = \mathrm{diag}(1, 2) \in G$ and a lift $\tilde{g} \in \mathrm{Gal}(L/\mathbf{Q})$. The elements $\tau\tilde{g}$ with $\tau \in \mathrm{Gal}(L/K)$ are the lifts of g, and Φ of the cocycle at $\tau\tilde{g}$ is $\Phi(\tau) + \Phi(\tilde{g})$, so as τ varies the vector $(t_1, \dots, t_r) = (\kappa(P_i)(\tau\tilde{g}))_i$ runs through all of $E[p]^r$. Let $\ell \nmid pN$ be a prime unramified in L whose Frobenius is $\tau\tilde{g}$. Its image g fixes exactly a line of $E[p]$, so $\tilde{E}(\mathbf{F}_\ell)[p] \cong \mathbf{Z}/p$ and the p-Sylow subgroup of $\tilde{E}(\mathbf{F}_\ell)$ is cyclic and nontrivial. Moreover $\tilde{P}_i \in p\tilde{E}(\mathbf{F}_\ell)$ if and only if some $Q' \in Q_i + E[p]$ is fixed by Frobenius, that is, if and only if $t_i \in (g-1)E[p]$. Since $(g-1)E[p]$ is a line, $E[p]/(g-1)E[p] \cong \mathbf{F}_p$, and $\phi_\ell(\tilde{P}_i)$ is, up to a nonzero scalar depending on ℓ, the image $\psi(t_i)$ of t_i in this quotient. Choosing τ so that $(\psi(t_1), \dots, \psi(t_r))$ is the kth unit vector, for $k = 1, \dots, r$, and applying the Chebotarev density theorem in $L/\mathbf{Q}$ to each of these r classes gives infinitely many choices of $\mathcal{L}$ whose vectors span $\mathbf{F}_p^r$. □

In practice $\mathcal{L}$ is assembled from the primes ℓ with $p \mid \#\tilde{E}(\mathbf{F}_\ell)$, keeping a prime when its vector is independent of those already kept; in every case treated in this paper a set of exactly r primes was found. Since such primes have density close to $1/p$, the size of the primes needed is expected to grow like $p \log p$; the largest, $\ell = 102\,337$, occurs for E_4 at $p = 823$, and for the rank-three pairs all of them are below 10^4.

8. The rank-four curve

8.1. **The curve.** The model $y^2 + xy = x^3 - x^2 - 79x + 289$ is minimal, with discriminant $\Delta = 2^2 \cdot 117\,223 > 0$ and conductor $2 \cdot 117\,223$; the reduction is of type I_2 at 2 and I_1 at $117\,223$, so $c_2 = 2$, $c_{117223} = 1$ and $\prod c_v = 2$. Since $\Delta > 0$, $E_4(\mathbf{R})$ has two components and $\Omega_{E_4} = 2\Omega_0$. The torsion subgroup injects into $\tilde{E}_4(\mathbf{F}_3)$ and into $\tilde{E}_4(\mathbf{F}_5)$, of orders 7 and 10, so $E_4(\mathbf{Q})_{\mathrm{tors}} = 0$. Irreducibility witnesses for the twelve primes of Mazur's list are

$$\ell = 2, 5, 11, 17\colon q = 3; \qquad \ell = 3, 7, 19, 43, 67, 163\colon q = 5; \qquad \ell = 13, 37\colon q = 11,$$

so the isogeny class of E_4 is $\{E_4\}$. The four points

$$P_1 = (6, -1), \quad P_2 = (4, 3), \quad P_3 = (5, -2), \quad P_4 = (8, 7)$$

lie on E_4. Their independence follows from the computation below, since $\mathrm{Reg}_5(P_1, \dots, P_4) \neq 0$.

8.2. **The prime five by hand.** We carry out every step of Theorem 5.1 for $p = 5$. Here $a_5 = -4$, so 5 is good ordinary, and $\#\tilde{E}_4(\mathbf{F}_5) = 10$: the prime is anomalous.

Surjectivity. The traces $a_3 = -3$, $a_{13} = -6$ give $a_3^2 - 12 = -3 \equiv 2$, a nonsquare modulo 5, and $a_{13}^2 - 52 = -16 \equiv 4$, a nonzero square; and $u = a_3^2/3 = 3$ modulo 5 satisfies $u \notin \{0, 1, 2, 4\}$ and $u^2 - 3u + 1 = 1$. By Proposition 6.1, $\bar{\rho}_{E_4,5}$ is surjective.

The coefficient a_4. For $p = 5$ and $k = 4$ only $j = 4$ occurs in (6). Since $a_5 \equiv 1 \pmod 5$,

$$a_4 \equiv \sum_{a=1}^{4} \Big(\Big[\frac{b_{a,4}}{25} \Big]^+ - \Big[\frac{a}{5} \Big]^+ \Big) \pmod 5, \qquad b_{a,4} \equiv a^5 \cdot 6^4 \pmod{25}.$$

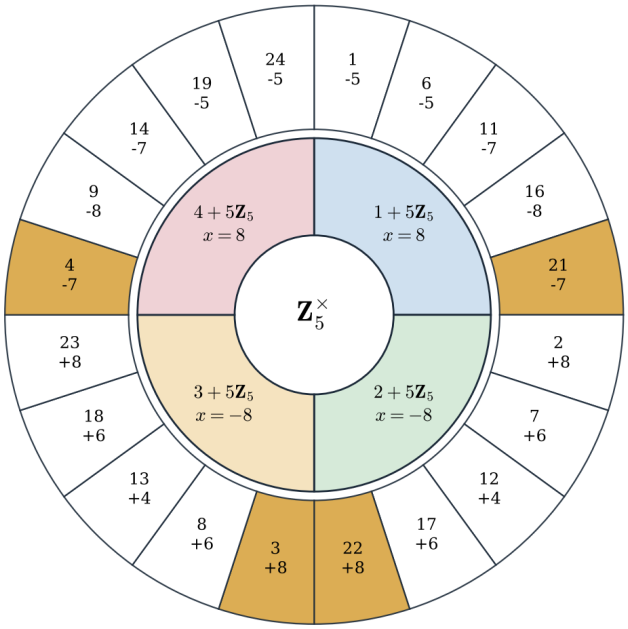


FIGURE 4. The classes $b + 25\mathbf{Z}_5$ of $\mathbf{Z}_5^\times$, grouped by $b \bmod 5$, for E_4. Each outer cell shows b and $2[b/25]^+$; each inner cell shows $2[a/5]^+$. The four shaded classes $\omega(a) \cdot 6^4$ give the coefficient of T^4.

The fifth powers of $1, 2, 3, 4$ modulo 25 are $1, 7, 18, 24$ (the Teichmüller representatives), and $6^4 \equiv 21$, so $b_{a,4} = 21, 22, 3, 4$. The eight modular symbols needed are

a	1	2	3	4
$[a/5]^+$	4	-4	-4	4
$b_{a,4}$	21	22	3	4
$[b_{a,4}/25]^+$	$-\frac{7}{2}$	4	4	$-\frac{7}{2}$

and therefore

$$a_4 \equiv \left(-\tfrac{7}{2} - 4\right) + (4 + 4) + (4 + 4) + \left(-\tfrac{7}{2} - 4\right) = 1 \pmod 5.$$

Figure 4 shows all twenty classes modulo 25 with their symbols. Using all of them in (6) gives $a_0 \equiv a_1 \equiv a_2 \equiv a_3 \equiv 0 \pmod 5$, as Remark 5.2 predicts.

The regulator. The unit root is $\alpha = 1 + 3 \cdot 5 + 2 \cdot 5^2 + O(5^3)$, so $1 - \alpha^{-1} = (\alpha - 1)/\alpha \equiv 3 \cdot 5 \pmod{5^2}$ and

$$(1 - \alpha^{-1})^2 = 4 \cdot 5^2 + 4 \cdot 5^3 + O(5^4).$$

The 5-adic regulator of the four points and the logarithm of the generator are

$$\mathrm{Reg}_5(P_1, \dots, P_4) = 2 \cdot 5^2 + 2 \cdot 5^3 + O(5^5), \qquad \log_5(6) = 5 + 2 \cdot 5^2 + 4 \cdot 5^3 + O(5^4),$$

so $\mathrm{Reg}_\gamma = \mathrm{Reg}_5 / \log_5(6)^4 = 2 \cdot 5^{-2} + 5^{-1} + O(1)$, and

$$S_5 = (1 - \alpha^{-1})^2 \cdot 2 \cdot \mathrm{Reg}_\gamma = 1 + 2 \cdot 5 + O(5^2).$$

The factor 5^2 coming from the anomalous reduction is cancelled exactly by the denominator 5^{-2} of the normalised regulator.

Saturation. With $\mathcal{L} = \{13, 23, 59, 67\}$ Proposition 7.1 applies; the vectors are computed in Appendix E.

Conclusion. $v_5(a_4) = 0 = v_5(S_5)$, so Theorem 5.1 gives $\operatorname{rank} E_4(\mathbf{Q}) = 4$, $Ш(E_4/\mathbf{Q})[5^\infty] = 0$, $\operatorname{ord}_T L_5(E_4, T) = 4$, $\operatorname{Reg}_5(E_4/\mathbf{Q}) \neq 0$, and the main conjecture for $(E_4, 5)$. As a further check, $a_4 \equiv S_5 \pmod 5$, which is what the p-adic leading-term conjecture predicts if $Ш(E_4/\mathbf{Q})$ is trivial.

8.3. **The other ordinary primes.** The same steps were carried out for each of the 163 good ordinary primes $5 \leq p < 1000$; the supersingular primes in this range are 59, 101 and 337. For each p, a surjectivity witness (q_1, q_2, q_3) was found (Table 2), the points were shown to be p-saturated by Proposition 7.1, and a_4 was computed from (6). For 161 of the primes, $v_p(S_p) = 0$ and $a_4 \not\equiv 0 \pmod p$. The remaining two, $p = 7$ and $p = 157$, are irregular: $v_p(\operatorname{Reg}_\gamma) = 1$ while p is not anomalous, so $v_p(S_p) = 1$. There Proposition 2.3 with $n = 3$ gives

$$a_4 \equiv 3 \cdot 7 \pmod{7^2}, \qquad a_4 \equiv 34 \cdot 157 \pmod{157^2},$$

so $v_p(a_4) = 1 = v_p(S_p)$ in both cases. At each of the 161 primes with $v_p(S_p) = 0$ the residue of a_4/S_p modulo p is 1. Theorem 5.1 applies at all of them.

The prime 7 shows what an irregular prime looks like in practice. Here $a_7 = -5$, so $\#\tilde{E}_4(\mathbf{F}_7) = 13$ and 7 is not anomalous, and $\prod c_v = 2$ is a unit. The unit root and the Euler factor are

$$\alpha = 2 + 2 \cdot 7 + 3 \cdot 7^3 + O(7^4), \qquad (1 - \alpha^{-1})^2 = 2 + 2 \cdot 7 + 3 \cdot 7^2 + O(7^3),$$

both units. The regulator, on the other hand, is divisible by 7^5 rather than by the 7^4 that the normalisation removes:

$$\operatorname{Reg}_7(P_1, \dots, P_4) = 6 \cdot 7^5 + 4 \cdot 7^6 + O(7^7), \qquad \log_7(8) = 7 + 3 \cdot 7^2 + O(7^3),$$

so $\operatorname{Reg}_\gamma = 6 \cdot 7 + 2 \cdot 7^2 + O(7^3)$ and $S_7 = 3 \cdot 7 + 0 \cdot 7^2 + O(7^3)$. At the second level the Riemann sum gives $a_{2,4} \equiv 0 \pmod 7$, which by Proposition 2.3 only says $v_7(a_4) \geq 1$ and decides nothing. At the third level, a sum over the $7^2 \cdot 6 = 294$ classes of $\mathbf{Z}_7^\times$ modulo 7^3, or 147 after the symmetry of Proposition 2.4, it gives $a_4 \equiv 3 \cdot 7 \pmod{7^2}$. So $v_7(a_4) = 1 = v_7(S_7)$, and the two leading digits agree as well. The witness is $(5, 3, 3)$, and $\mathcal{L} = \{3, 19, 37, 79\}$ proves saturation.

8.4. **The supersingular primes.** At a supersingular prime there is no unit root, and the construction of §2 breaks down: both roots of $X^2 - a_p X + p$ have valuation $\frac{1}{2}$ (here $a_p = 0$, as $p \geq 5$ and $|a_p| < 2\sqrt{p}$ force $a_p = 0$ when $p \mid a_p$), and the measure (3) with either root is unbounded. Perrin-Riou's remedy is to work in the Dieudonné module $D_p(E) = H^1_{\mathrm{dR}}(E/\mathbf{Q}_p)$, a two-dimensional $\mathbf{Q}_p$-vector space with basis $\omega, \varphi(\omega)$, where φ is Frobenius, normalised to have characteristic polynomial $X^2 - p^{-1} a_p X + p^{-1}$. Let $L_{p,\alpha}(E, T)$ be the series of (4) built with a root α in place of the unit root, a power series with coefficients in $\mathbf{Q}_p(\alpha)$, and write $L_{p,\alpha} = G + \alpha H$ with $G, H \in \mathbf{Q}_p[[T]]$. Following Perrin-Riou [38, §3.5] put

$$L_p(E, T) = \big(G(T) + a_p H(T)\big)\, \omega - p\, H(T)\, \varphi(\omega),$$

a power series with coefficients in $D_p(E)$, which carries the same information as $L_{p,\alpha}$ and whose coordinates in the basis $\omega, \varphi(\omega)$ do not depend on the Weierstrass model. The p-adic height takes values in $D_p(E)$ as well, and so does the regulator $\operatorname{Reg}_p(E/\mathbf{Q})$. Perrin-Riou's analogue of Theorem 4.2 and Kato's divisibility combine exactly as in Theorem 5.1 [38, Thm. 6.2 and Algorithm 11.1]: if the leading coefficients are compared coordinate by coordinate, the bound is

$$v_p \#Ш(E/\mathbf{Q})[p^\infty] \leq \operatorname{ord}_p\big((1 - \varphi)^{-2} L_p^*(E, 0)\big) - \operatorname{ord}_p\Big(\prod_v c_v \cdot \operatorname{Reg}_\gamma(E/\mathbf{Q})\Big),$$

where $L_p^*(E,0)$ is the coefficient of T^r and ord_p of an element of $D_p(E)$ is the least valuation of its coordinates.

For $p = 59$ and $p = 101$ we computed both sides, using the basis $(-9,19)$, $(-8,23)$, $(-7,25)$, $(4,-7)$ of $E_4(\mathbf{Q})$, which is p-saturated at 59 and 101 by Proposition 7.1. At $p = 59$ the coordinates are

$$(1-\varphi)^{-2}L_{59}^*(E_4,0) = \big(22 + O(59),\ 54\cdot 59 + O(59^2)\big),$$
$$2\,\mathrm{Reg}_\gamma = \big(22 + 48\cdot 59 + O(59^2),\ 54\cdot 59 + 36\cdot 59^2 + O(59^3)\big),$$

and at $p = 101$

$$(1-\varphi)^{-2}L_{101}^*(E_4,0) = \big(61 + O(101),\ 36\cdot 101 + O(101^2)\big),$$
$$2\,\mathrm{Reg}_\gamma = \big(61 + 66\cdot 101 + O(101^2),\ 36\cdot 101 + 14\cdot 101^2 + O(101^3)\big).$$

The L-values come from the Riemann sums of level 3, in which the coefficients of $T^0,\dots,T^3$ vanish to the available precision. In both cases both sides have $\mathrm{ord}_p = 0$, so the bound is 0 and $Ш(E_4/\mathbf{Q})[59^\infty] = Ш(E_4/\mathbf{Q})[101^\infty] = 0$. As at the ordinary primes, the coordinates agree beyond their valuations: the first digits coincide in all four. The witnesses $(3,11,3)$ and $(3,5,3)$ show surjectivity at 59 and 101.

8.5. **The prime two.** A complete 2-descent on E_4 [9, §3.6] gives $\dim_{\mathbf{F}_2}\mathrm{Sel}^{(2)}(E_4/\mathbf{Q}) = 4$. Since $E_4(\mathbf{Q})[2] = 0$ and $\mathrm{rank}\,E_4(\mathbf{Q}) = 4$, the exact sequence $0 \to E_4(\mathbf{Q})/2E_4(\mathbf{Q}) \to \mathrm{Sel}^{(2)}(E_4/\mathbf{Q}) \to Ш(E_4/\mathbf{Q})[2] \to 0$ forces $Ш(E_4/\mathbf{Q})[2] = 0$. This completes the proof of Theorem A.

The primes 3 and 337 are not covered: $p = 3$ is excluded by the hypothesis $p \geq 5$ of Theorem 4.1 as used here, and at $p = 337$ the Riemann sums for the supersingular L-function were too large to evaluate.

9. Rank three

9.1. **The range.** Cremona's tables list 505 curves of rank three with conductor $30\,000 < N \leq 100\,000$, in 494 isogeny classes, together with three independent points on each. None has complex multiplication. For each class we take the first curve E and each prime $5 \leq p < 100$ of good ordinary reduction, 10 404 pairs in all. By Proposition 5.4 the conclusions transfer to the other curves of the class.

9.2. **Proof of Theorem B.** For each pair:

(1) a surjectivity witness exists, so $\bar\rho_{E,p}$ is surjective (no pair failed);
(2) the three points are p-saturated, by Proposition 7.1;
(3) S_p and a_3 were computed, a_3 by (6).

For 9 985 pairs, $v_p(S_p) = 0$ and $a_3 \not\equiv 0 \pmod p$, so (8) holds at the second level. The remaining 419 pairs all have $v_p(S_p) > 0$; for them the precision of the second level is not enough, and the Riemann sums of higher level were evaluated. By Proposition 2.3, a pair with $v_p(S_p) = v$ can be settled at level $v+2$ and no lower, since $a_{n,3}$ determines a_3 only modulo p^{n-1}. Level $v+2$ sufficed in every case: there are 374 pairs with $v = 1$, 42 with $v = 2$ and three with $v = 3$, and for each of them $v_p(a_3) = v$; the pairs with $v \geq 2$ are listed in Table 3. Figure 6 shows how the pairs are spread over primes and levels: the deeper levels are needed almost only at the small primes, where the sums are cheap. The most expensive pair is $(60803a1, 89)$, where $v_{89}(S_{89}) = 2$ and $a_3 \equiv 84\cdot 89^2 \pmod{89^3}$ required the sum $L_{89,4}$, with about $3.1\cdot 10^7$ modular symbols of denominator 89^4 after the symmetry $[-r]^+ = [r]^+$ is used. Thus (8) holds

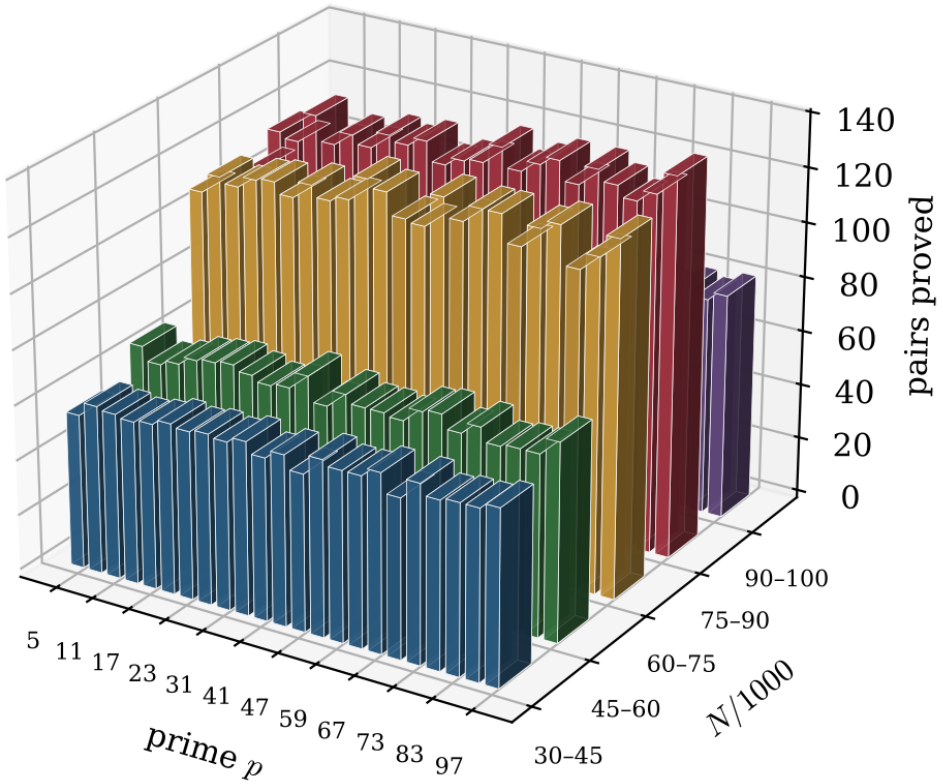


FIGURE 5. Pairs (E, p) settled in Theorem B, by prime and by conductor.

for all 10 404 pairs, and Theorem 5.1 gives Theorem B. At each of the 9 985 pairs settled at the second level, the residue of a_3/S_p modulo p is 1.

Since every class contains a pair to which Theorem 5.1 applies, $\operatorname{rank} E(\mathbf{Q}) = 3$ for every curve in the range.

9.3. **Two irregular pairs.** Two examples show how the deeper sums settle irregular pairs. In both, the Riemann sum of the second level returns $a_{2,3} \equiv 0 \pmod p$, which is consistent with any $v_p(a_3) \geq 1$ and decides nothing.

The curve 30446*a*1 *at* $p = 5$. The minimal model is $y^2 + xy = x^3 + x^2 - 39x - 59$, with conductor $2 \cdot 13 \cdot 1171$, discriminant $2^4 \cdot 13^2 \cdot 1171$, $\prod c_v = 4$ and trivial torsion; the points

$$P_1 = (-3, 8), \qquad P_2 = (10, 21), \qquad P_3 = (-5, 9)$$

are independent. Here $a_5 = -3$, $\#\tilde{E}(\mathbf{F}_5) = 9$, and

$$\alpha = 2 + 5 + 5^2 + 3 \cdot 5^3 + O(5^4), \qquad (1 - \alpha^{-1})^2 = 4 + 2 \cdot 5 + O(5^3).$$

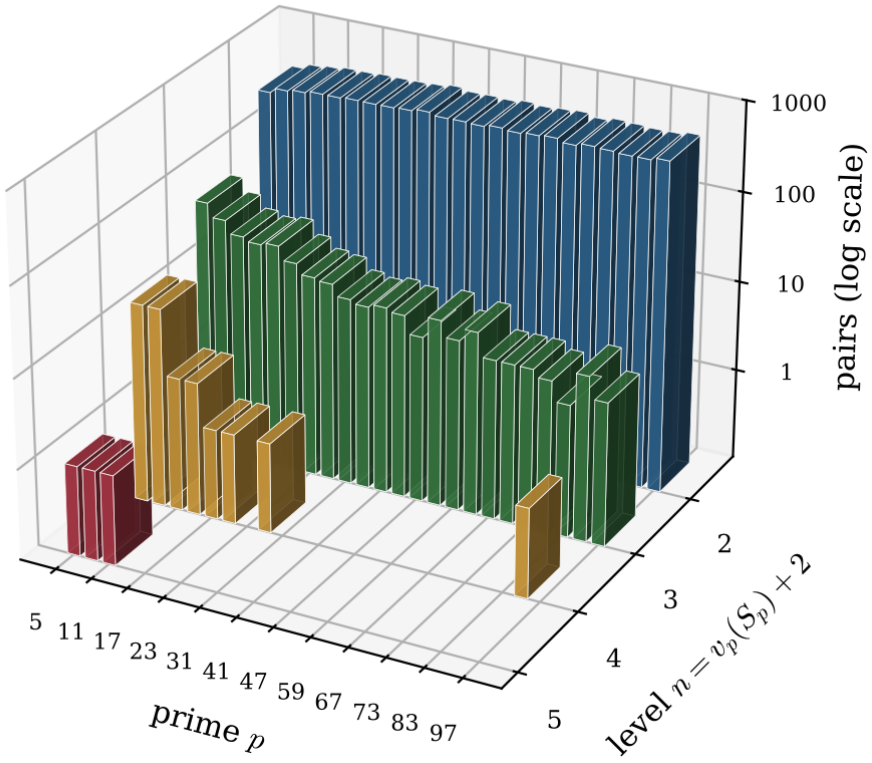


FIGURE 6. The level n of the Riemann sum at which each rank-three pair is settled, by prime, on a logarithmic scale. A pair with $v_p(S_p) = v$ is settled at level $v + 2$ and at no lower level.

Modulo 5^4 the height pairing on P_1, P_2, P_3 is

$$\big(\langle P_i, P_j\rangle_5\big) \equiv \begin{pmatrix} 300 & 175 & 30 \\ 175 & 325 & 450 \\ 30 & 450 & 30 \end{pmatrix} \pmod{5^4},$$

every entry divisible by 5. Since the entries are known to an error of 5^4 and have valuation at least 1, the determinant is known modulo 5^6, and it is $1250 = 2 \cdot 5^4$. So $\mathrm{Reg}_5 = 2 \cdot 5^4 + 0 \cdot 5^5 + O(5^6)$, and with $\log_5(6) = 5 + 2 \cdot 5^2 + 4 \cdot 5^3 + O(5^4)$,

$$\mathrm{Reg}_\gamma = \frac{\mathrm{Reg}_5}{\log_5(6)^3} = 2 \cdot 5 + 3 \cdot 5^2 + O(5^3), \qquad S_5 = (1 - \alpha^{-1})^2 \cdot 4 \cdot \mathrm{Reg}_\gamma = 2 \cdot 5 + 0 \cdot 5^2 + O(5^3).$$

The prime 5 is not anomalous and divides no Tamagawa number; the irregularity comes entirely from the regulator, which is divisible by 5^4 instead of 5^3. The sum of the third level runs over the 100 classes of $\mathbf{Z}_5^\times$ modulo 5^3, or 50 after the symmetry of Proposition 2.4, and gives $a_3 \equiv 2{\cdot}5 \pmod{5^2}$. So $v_5(a_3) = 1 = v_5(S_5)$, and the leading digits agree. With the witness $(17, 3, 17)$ and $\mathcal{L} = \{3, 11, 31\}$, Theorem 5.1 gives $Ш(E/\mathbf{Q})[5^\infty] = 0$.

The curve 60803a1 *at* $p = 89$. The minimal model is $y^2 + y = x^3 + x^2 - 65x + 195$, with conductor $41 \cdot 1483$, discriminant $-41^2 \cdot 1483$, $\prod c_v = 2$ and trivial torsion; the points are

$$P_1 = (-5, 20), \qquad P_2 = \left(\tfrac{21}{4}, \tfrac{37}{8}\right), \qquad P_3 = \left(\tfrac{37}{9}, \tfrac{89}{27}\right).$$

Here $a_{89} = -12$ and $\#\tilde{E}(\mathbf{F}_{89}) = 102$, so again only the regulator contributes:

$$\alpha = 77 + 51 \cdot 89 + O(89^2), \qquad (1 - \alpha^{-1})^2 = 50 + 47 \cdot 89 + O(89^2),$$
$$\mathrm{Reg}_{89} = 40 \cdot 89^5 + 41 \cdot 89^6 + O(89^7), \qquad \log_{89}(90) = 89 + 44 \cdot 89^2 + O(89^3),$$

so $\mathrm{Reg}_\gamma = 40 \cdot 89^2 + 12 \cdot 89^3 + O(89^4)$ and

$$S_{89} = 84 \cdot 89^2 + 20 \cdot 89^3 + O(89^4).$$

This is the only pair with $p \geq 29$ and $v_p(S_p) = 2$. By Proposition 2.3 the coefficient a_3 is determined modulo 89^{n-1} by the sum of level n, so level 4 is the first that can see $v_{89}(a_3) = 2$. The third level returns 0 modulo 89^2; the fourth needs $[b/89^4]^+$ for $88 \cdot 89^3/2 \approx 3.1 \cdot 10^7$ classes b and gives

$$a_3 \equiv 84 \cdot 89^2 \pmod{89^3}.$$

The witness is $(7, 3, 3)$ and $\mathcal{L} = \{79, 269, 983\}$, so Theorem 5.1 gives $Ш(E/\mathbf{Q})[89^\infty] = 0$ for every curve in the isogeny class 60803a.

In both examples the first nonzero digit of a_3 equals that of S_p. This is the congruence $a_r \equiv S_p$ modulo $p^{v_p(S_p)+1}$ which the p-adic leading-term formula predicts once $Ш(E/\mathbf{Q})[p^\infty] = 0$; it is not used in the proofs, but it holds for all 419 irregular rank-three pairs, as well as for E_4 at 7 and 157.

9.4. **How often a prime is irregular.** A pair is irregular when $v_p(S_p) > 0$. Three quantities contribute to $v_p(S_p)$: the Euler factor, of valuation $2v_p(\#\tilde{E}(\mathbf{F}_p))$; the Tamagawa product; and the normalised regulator Reg_γ, whose valuation can be negative. They are not independent. At an anomalous prime the Euler factor has valuation 2, but the regulator usually compensates: of the 320 anomalous pairs in the range, 285 are regular, with $v_p(\mathrm{Reg}_\gamma) = -2$, as for E_4 at 5. Remark 5.3 explains why the total is never negative once $Ш(E/\mathbf{Q})[p^\infty]$ is known to vanish. If the residues of these quantities modulo p behaved like random elements of $\mathbf{F}_p$, a fixed p would be irregular for a proportion of curves close to c/p for a constant c. Figure 7 shows p times the observed proportion, and Table 4 lists the counts. The product stays between 0.55 and 1.55 across the whole range, with mean 1.00, so the proportion of irregular pairs at p falls off like $1/p$, as the random model predicts. The irregular pairs are the expensive ones, since they need Riemann sums at the third or fourth level, and their number therefore drops quickly as p grows.

10. Ranks five and six

10.1. **The curves.** E_5 has prime conductor 19 047 851 and discriminant $-19\,047\,851$, so its reduction at the conductor is of type I_1, $\prod c_v = 1$ and $E_5(\mathbf{R})$ is connected. E_6 has discriminant $2^2 \cdot 3^2 \cdot 2777 \cdot 311\,341$, reduction of type I_2 at 2 and 3 and I_1 at the other two primes, $\prod c_v = 4$, and two real components. Both curves are semistable. Their torsion is trivial: for E_5 the orders $\#\tilde{E}_5(\mathbf{F}_3) = 7$ and $\#\tilde{E}_5(\mathbf{F}_5) = 9$ are coprime; for E_6 the orders $10, 12, 18, 20$ at $5, 7, 11, 13$ have greatest common divisor 2, and $E_6(\mathbf{Q})[2] = 0$ by the irreducibility witness $q = 17$, $a_{17} = -7$. Irreducibility witnesses exist for all twelve primes of Mazur's list, so neither curve has a rational isogeny; in particular each is the optimal curve of its class, and its Manin constant is 1 by Česnavičius' theorem for semistable curves [8].

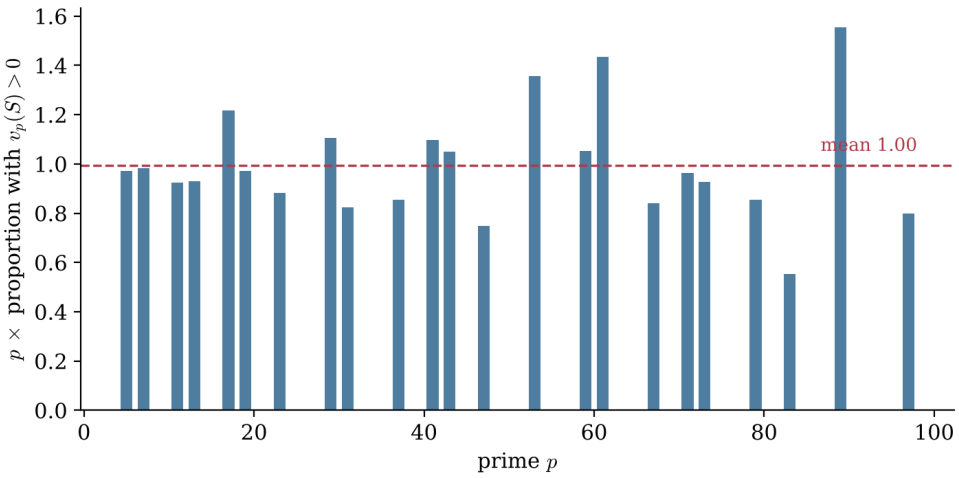


FIGURE 7. For each prime $p < 100$, p times the proportion of the rank-three pairs with $v_p(S_p) > 0$.

The points

$$E_5\colon\ (-10,11),\ (-\tfrac{39}{4},\tfrac{105}{8}),\ (-8,21),\ (-7,23),\ (-6,24),$$
$$E_6\colon\ (-59,35),\ (-58,116),\ (-\tfrac{229}{4},\tfrac{1141}{8}),\ (-53,227),\ (-\tfrac{193}{4},\tfrac{2215}{8}),\ (32,-4)$$

lie on the curves and are p-saturated at every prime $p < 100$ by Proposition 7.1.

10.2. **Modular symbols at large conductor.** At these conductors the space of modular symbols is too large for the linear-algebra method of [9]. The values $[r]^+$ were instead computed by Wuthrich's numerical method [40]: the integral $\lambda^+(r)$ is evaluated from the Fourier expansion of f at a suitable cusp, with a rigorous bound for the error, and rounded to the unique rational number with the known denominator. The denominator bound is the one in the proof of Lemma 2.1(c), with $c = 1$; this is where the value of the Manin constant enters.[7]

10.3. **The prime seven for E_5.** As for E_4 at 5, every step can be followed by hand. Here $a_7 = -4$, so 7 is good ordinary, $\#\tilde{E}_5(\mathbf{F}_7) = 12$, and 7 divides no Tamagawa number. By Lemma 3.2 the regulator Reg_γ is a 7-adic integer, and the unit root of $X^2 + 4X + 7$ is

$$\alpha = 3 + 7 + 6\cdot 7^2 + O(7^3), \qquad 1 - \alpha^{-1} \equiv 1 - 5 \equiv 3 \pmod 7,$$

so $(1-\alpha^{-1})^2$ is a unit. With the five points of the previous subsection,

$$\mathrm{Reg}_7 = 2\cdot 7^5 + 6\cdot 7^6 + 4\cdot 7^7 + O(7^8), \qquad S_7(E_5) = (1-\alpha^{-1})^2\,\mathrm{Reg}_\gamma = 4 + 5\cdot 7 + 4\cdot 7^2 + O(7^3).$$

Every entry of the 5×5 height matrix is divisible by 7, as Lemma 3.2 predicts, and its determinant is divisible by 7^5 and by no higher power: the pair is regular.

[7] This is the only place where the Manin constant itself, and not merely its integrality, is used; [40] states the requirement explicitly.

For the Riemann sum, (6) with $k = 5$ involves only $j = 5$ and $j = 6$, with $\binom{5}{5} = 1$ and $\binom{6}{5} = 6$. All six symbols $[a/7]^+$ vanish, so only the symbols of denominator 49 contribute:

a	1	2	3	4	5	6
$b_{a,5}$	36	2	38	11	47	13
$[b_{a,5}/49]^+$	1	-1	0	0	-1	1
$b_{a,6}$	43	16	10	39	33	6
$[b_{a,6}/49]^+$	1	1	1	1	1	1

The symmetry $[-r]^+ = [r]^+$ is visible in the pairs $b + b' = 49$. Since $a_7 \equiv 3$ and $3^{-2} \equiv 4 \pmod 7$,

$$a_5 \equiv 4\big((1 - 1 + 0 + 0 - 1 + 1) + 6 \cdot (1 + 1 + 1 + 1 + 1 + 1)\big) = 144 \equiv 4 \pmod 7,$$

which is the first digit of $S_7(E_5)$. The same computation with $k \le 4$ gives $a_0 \equiv \cdots \equiv a_4 \equiv 0$. So $v_7(a_5) = v_7(S_7) = 0$, and with the witness $(2, 3, 3)$ Theorem 5.1 gives $\operatorname{rank} E_5(\mathbf{Q}) = 5$ and $Ш(E_5/\mathbf{Q})[7^\infty] = 0$.

One level higher the agreement persists. Proposition 2.4 with $n = 3$ uses the 168 symbols of denominators 49 and 343 that remain after the symmetry, and gives

$$a_0 \equiv \cdots \equiv a_4 \equiv 0, \qquad a_5 \equiv 39 = 4 + 5 \cdot 7 \pmod{7^2},$$

so the first two digits of a_5 and of $S_7(E_5)$ coincide. This is what the p-adic conjecture of Mazur, Tate and Teitelbaum predicts once $Ш(E_5/\mathbf{Q})[7^\infty] = 0$ is known (Section 11).

10.4. **Proof of Theorems C and D.** For E_5 and $p = 7, 11, 13, 17, 19$ all pairs are regular, $v_p(S_p) = 0$, and (6) gives

$$a_5 \equiv 4,\ 2,\ 7,\ 13,\ 16 \pmod p \quad \text{for } p = 7, 11, 13, 17, 19.$$

The surjectivity witnesses are $(2, 3, 3)$, $(2, 17, 3)$, $(5, 2, 3)$, $(3, 2, 3)$, $(2, 3, 3)$. Theorem 5.1 applies at each of these primes.

For E_6 and $p = 7$, the witness $(5, 19, 5)$ shows surjectivity, $v_7(S_7) = 0$, and (6) with $k = 6$ gives $a_6 \equiv 6 \pmod 7$, with $a_0, \ldots, a_5 \equiv 0$. Theorem 5.1 applies. In all six cases $a_r \equiv S_p \pmod p$.

The primes 5 for E_5 (where $v_5(S_5) = 1$) and $p \ge 11$ for E_6, and $p \ge 23$ for E_5, are left open: the Riemann sums at these primes require modular symbols with denominators p^2 or p^3, and their evaluation at these conductors exceeded the computation we could carry out.

11. The leading coefficient

Theorem 5.1 compares the valuations of two numbers, the coefficient a_r of T^r in $L_p(E, T)$ and

$$S_p(E) = \Big(1 - \frac{1}{\alpha}\Big)^2 \frac{\prod_v c_v \cdot \operatorname{Reg}_\gamma(E/\mathbf{Q})}{\#E(\mathbf{Q})^2_{\mathrm{tors}}},$$

and it uses nothing beyond the valuations. The p-adic conjecture of Mazur, Tate and Teitelbaum [27], in the form of [38] for good ordinary p, predicts the numbers themselves: if $Ш(E/\mathbf{Q})$ is finite, then

$$a_r = \#Ш(E/\mathbf{Q}) \cdot S_p(E). \tag{9}$$

Theorem 5.1 proves the p-part of this identity, $v_p(a_r) = v_p(S_p)$ together with $v_p \#Ш(E/\mathbf{Q}) = 0$. The remaining content of (9) at a pair where $Ш[p^\infty] = 0$ is a congruence between a_r and S_p beyond their common valuation, and it can be tested by the same Riemann sums one level higher.

Proposition 11.1. *Let (E,p) be one of the 1 717 pairs of Theorem B with $p \in \{5,7,11,13\}$, and let $v = v_p(S_p(E))$. Then*

$$a_3 \equiv S_p(E) \pmod{p^{v+2}}.$$

Consequently, if $Ш(E/\mathbf{Q})$ is finite, then (9) holds modulo p^{v+2} if and only if $\#Ш(E/\mathbf{Q}) \equiv 1 \pmod{p^2}$.

Proof. For each pair, Proposition 2.4 with $n = v+3$ gives a_3 modulo p^{v+2}, since $3 < p$; the symbols needed have denominators p^{v+2} and p^{v+3}. The regulator is computed to p-adic precision p^{v+8} by the method of §3.1; after division by $\log_p(1+p)^3$, of valuation 3, this determines S_p modulo p^{v+2} with room to spare. The two residues coincide in every case. For the second statement write $S_p = p^v u$ with $u \in \mathbf{Z}_p^\times$; then $\#Ш \cdot S_p - a_3 \equiv (\#Ш - 1)p^v u \pmod{p^{v+2}}$, which vanishes modulo p^{v+2} exactly when $p^2 \mid \#Ш - 1$. □

The congruence of Proposition 11.1 is an independent test: Theorem 5.1 only needed the first p-adic digits of a_3 and S_p to be nonzero, while the proposition matches their first two digits. If the second digits were unrelated, each pair would pass with probability $1/p$, and the 1 717 pairs together would pass with probability below 5^{-1717}. Figure 8 shows the second digits at $p = 7$ and $p = 13$: they are spread over all residues, and every pair sits on the diagonal. Table 1 lists the pairs with $v \geq 2$, where the congruence holds modulo p^4, and in three cases modulo p^5.

The classical side gives the expected value of $\#Ш$. For the first curve of each of the 8 862 isogeny classes of rank three with conductor between 30 000 and 500 000 in Cremona's tables, the number

$$\#Ш_{\mathrm{an}}(E/\mathbf{Q}) = \frac{L^{(3)}(E,1)}{3!} \cdot \frac{\#E(\mathbf{Q})^2_{\mathrm{tors}}}{\Omega_E \prod_v c_v \mathrm{Reg}(E/\mathbf{Q})},$$

computed from the generators in the tables with the real height pairing, is 1 to within 10^{-15}; the same holds for E_4, E_5 and E_6 with the derivatives of order four, five and six.[8] So the classical conjecture predicts $\#Ш(E/\mathbf{Q}) = 1$, (9) becomes $a_r = S_p$, and Proposition 11.1 is this identity modulo p^{v+2}, proved for every pair it covers. For E_5 at 7 the first two digits of a_5 and S_7 agree by §10.3, and for E_4 at its two irregular primes 7 and 157 the first digits agree by the computation at level three in §8.

Two remarks put this in context. First, (9) is not known for a single curve of rank at least two, since it presupposes the finiteness of $Ш(E/\mathbf{Q})$; what can be proved, here and in [38], is its p-part at explicit pairs. Second, the proposition does not depend on $\#Ш_{\mathrm{an}}$, which enters only in its interpretation; what is proved is the congruence between the two explicit p-adic numbers a_3 and S_p.

12. From the p-parts to the conjecture

Theorem 5.1 treats one prime at a time and says nothing about $L(E,s)$ at $s = 1$. This section sets out precisely what separates the results of this paper from the conjecture (1) for the same curves, because the missing pieces are of different kinds: one concerns the complex L-function, one concerns infinitely many primes, and one concerns the comparison of the p-adic and complex sides.

[8] This is a numerical computation of $L^{(r)}(E,1)$ by the rapidly converging series of the functional equation; it is not a proof that $L^{(r)}(E,1)$ has the computed value, and none of the theorems of this paper depends on it.

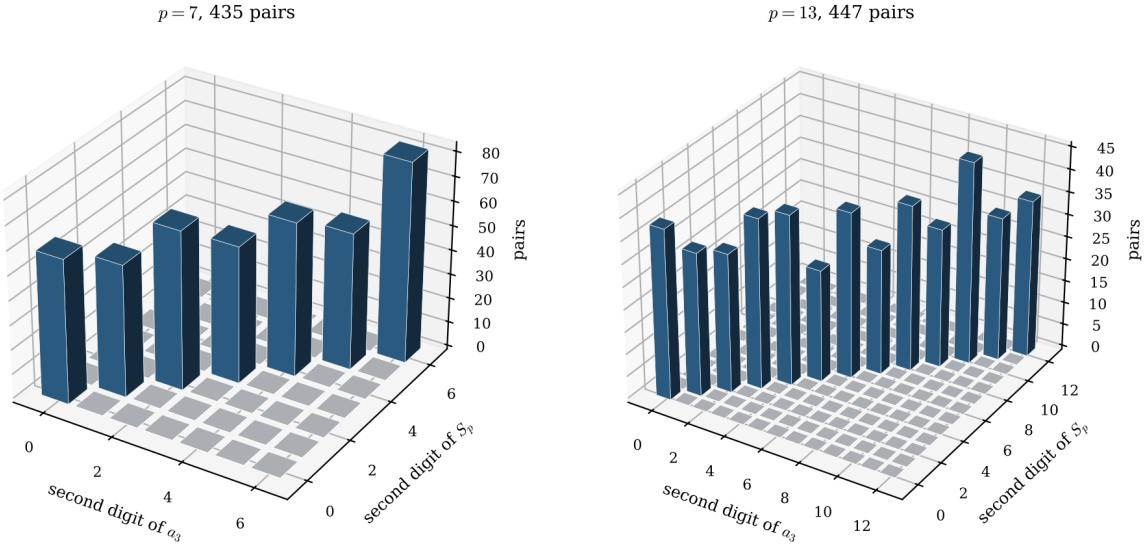


FIGURE 8. The digit of p^{v+1} in a_3 and in S_p, where $v = v_p(S_p)$, for the pairs of Proposition 11.1 at $p = 7$ and $p = 13$. Grey tiles are empty cells; all pairs lie on the diagonal.

TABLE 1. The pairs of Proposition 11.1 with $v_p(S_p) \geq 2$: the common residue of a_3 and S_p modulo p^{v+2}, written in base p from the digit of p^v.

curve	p	v	$a_3 \equiv S_p$	curve	p	v	$a_3 \equiv S_p$
33609a1	11	2	$10 \cdot 11^2 + 2 \cdot 11^3$	73931a1	7	2	$5 \cdot 7^2$
38600m1	7	2	$7^2 + 7^3$	74216a1	5	2	$4 \cdot 5^2 + 3 \cdot 5^3$
43556b1	5	2	$5^2 + 5^3$	74434a1	5	2	$2 \cdot 5^2 + 4 \cdot 5^3$
45352a1	5	2	$3 \cdot 5^2 + 2 \cdot 5^3$	74864b1	7	2	$2 \cdot 7^2 + 4 \cdot 7^3$
47087a1	5	2	$4 \cdot 5^2 + 4 \cdot 5^3$	78287b1	7	2	$2 \cdot 7^2 + 4 \cdot 7^3$
47696h1	5	2	$2 \cdot 5^2$	80256a1	5	2	$4 \cdot 5^2 + 4 \cdot 5^3$
49411a1	5	2	$3 \cdot 5^2 + 3 \cdot 5^3$	81328c1	7	2	$4 \cdot 7^2 + 6 \cdot 7^3$
49626a1	7	2	$4 \cdot 7^2 + 3 \cdot 7^3$	83928b1	5	2	$3 \cdot 5^2 + 3 \cdot 5^3$
51052a1	7	2	$4 \cdot 7^2 + 7^3$	87866a1	5	2	$5^2 + 2 \cdot 5^3$
53122a1	5	3	$5^3 + 2 \cdot 5^4$	88022a1	7	2	$4 \cdot 7^2 + 5 \cdot 7^3$
53122a1	7	2	$3 \cdot 7^2 + 6 \cdot 7^3$	88502f1	5	2	$3 \cdot 5^2 + 2 \cdot 5^3$
53957b1	5	2	$2 \cdot 5^2 + 5^3$	88883a1	7	2	$5 \cdot 7^2 + 6 \cdot 7^3$
54489b1	7	2	$4 \cdot 7^2 + 2 \cdot 7^3$	89294a1	11	3	$5 \cdot 11^3 + 3 \cdot 11^4$
59450i1	7	2	$5 \cdot 7^2 + 2 \cdot 7^3$	90953a1	7	3	7^3
61886d1	7	2	$4 \cdot 7^2 + 7^3$	91506a1	5	2	$2 \cdot 5^2 + 2 \cdot 5^3$
62576b1	13	2	$12 \cdot 13^2$	91958a1	5	2	$4 \cdot 5^2 + 3 \cdot 5^3$
63574d1	5	2	$2 \cdot 5^2$	92651a1	11	2	$5 \cdot 11^2 + 10 \cdot 11^3$
65122e1	7	2	$5 \cdot 7^2 + 7^3$	94281b1	5	2	$2 \cdot 5^2 + 4 \cdot 5^3$
67064a1	11	2	$6 \cdot 11^2$	97111a1	13	2	$2 \cdot 13^2 + 13^3$
70449b1	7	2	$6 \cdot 7^2$	98231a1	13	2	$6 \cdot 13^2 + 10 \cdot 13^3$
71688a1	7	2	$7^2 + 6 \cdot 7^3$				

12.1. **The analytic rank.** The order of vanishing of $L(E, s)$ at $s = 1$ can be bounded above by a computation: a single value $L^{(k)}(E, 1)$ shown to be nonzero with a rigorous error bound

gives $\operatorname{ord}_{s=1} L(E,s) \le k$. Bounding it below is harder, since a vanishing derivative cannot be recognised numerically. Two exact tools are available. The global root number $w(E) = \pm 1$, the sign in the functional equation of $L(E,s)$, equals $(-1)^{\operatorname{ord}_{s=1} L(E,s)}$ and is computed exactly as a product of local root numbers [9]. And the theorems of Gross–Zagier and Kolyvagin determine the rank when the analytic rank is at most one. Combined with the results of this paper they give a lower bound.

Corollary 12.1. *For every curve E of Theorem B and for E_5, $\operatorname{ord}_{s=1} L(E,s)$ is odd and at least 3. For E_4 and E_6 it is even and at least 2.*

Proof. Let $m = \operatorname{ord}_{s=1} L(E,s)$. If $m \le 1$, then $\operatorname{rank} E(\mathbf{Q}) = m$ by the formula of Gross and Zagier [15] and Kolyvagin's Euler system [19], with the auxiliary imaginary quadratic field supplied by the nonvanishing theorems of Bump, Friedberg and Hoffstein [2] and of Murty and Murty [23]. Theorems A–D give $\operatorname{rank} E(\mathbf{Q}) \ge 3$ in every case, so $m \ge 2$. The root number is -1 for every curve of Theorem B and for E_5, and $+1$ for E_4 and E_6; so m is odd in the first case and even in the second. □

For the curves of rank three the corollary leaves only the upper bound, which is a matter of showing $L'''(E,1) \ne 0$; Buhler, Gross and Zagier [3] did this with rigorous error control for the curve $5077a1$, the smallest conductor of rank three, and the same evaluation applies to any curve once $L'''(E,1)$ is computed with a certified error. For rank four and higher there is no comparable device. One would need to show $L''(E_4,1) = 0$, and no formula expresses $L''(E,1)$ as the height of an algebraic object, so the analytic rank of E_4 is known only to be even and at least 2, while a numerical evaluation gives $L^{(4)}(E_4,1) \approx 214.65$, which with a certified error bound would give the upper bound 4.

12.2. **An upper bound for the Selmer group from $L^{(r)}(E,1)$.** For $r \le 1$ the Euler system of Heegner points bounds the Selmer group [19], because by the formula of Gross and Zagier the basic Heegner point is of infinite order exactly when $L'(E/K,1) \ne 0$. The argument proves finiteness of Ш for all primes at once. For $r \ge 2$ there is no point whose height is $L^{(r)}(E,1)$. Over function fields Yun and Zhang [41] constructed such objects from moduli spaces of shtukas with r legs, and their higher Gross–Zagier formula expresses $L^{(r)}$ as an intersection number; no analogue over $\mathbf{Q}$ is known. Kolyvagin systems of higher rank exist [24], but a theorem that their leading class is nonzero when $L^{(r)}(E,1) \ne 0$ is missing.

The method of this paper replaces the complex derivative by the p-adic one, and Kato's Euler system supplies the bound that Heegner points supply in rank one. The price is that the p-adic statement must be checked one prime at a time: Theorem 5.1 needs $v_p(a_r)$, which is a property of the particular prime.

12.3. **Infinitely many primes.** To deduce the finiteness of $Ш(E/\mathbf{Q})$ from results of the kind proved here one would need Theorem 5.1 at all but finitely many primes, together with a separate argument at the remaining ones. Three obstacles appear. First, the hypothesis $\bar\rho_{E,p}$ surjective fails at finitely many primes by Serre's theorem [35], and at those the divisibility in Theorem 4.1 holds only up to a power of p. Second, half of the primes, in a sense made precise by the Sato–Tate law, are close to supersingular in the relevant range, and the supersingular theory of §8 is harder to evaluate. Third, and most seriously, the equality $v_p(a_r) = v_p(S_p)$ is a numerical coincidence at each p for which no uniform reason is known. The data of §9 show that it held without exception, which is what the p-adic Birch–Swinnerton-Dyer conjecture predicts if $Ш(E/\mathbf{Q})$ is trivial, but a proof for all p would be a proof of that conjecture for E.

12.4. **Schneider's conjecture.** Theorem 5.1 proves $\mathrm{Reg}_p(E/\mathbf{Q}) \neq 0$ for each pair it covers, but no argument gives it for all p. One route would be a transcendence statement for values of the p-adic sigma function at algebraic points, of the kind Bertrand obtained for curves with complex multiplication in rank one. For curves without complex multiplication, and for $r \geq 2$, no such statement is known: the regulator is a determinant of p-adic logarithms of algebraic points in a formal group, and a vanishing determinant would be an algebraic relation among them.

12.5. **The same order of vanishing for L_p and L.** Even if all p-adic statements were known, passing from $\mathrm{ord}_T\, L_p(E,T) = r$ to $\mathrm{ord}_{s=1}\, L(E,s) = r$ requires a link between derivatives of the two functions. The interpolation property of Proposition 2.5 relates L_p to the values $L(E,\chi,1)$ at finite-order characters, which are central values and not derivatives. For $r = 1$ the link is Perrin-Riou's p-adic Gross–Zagier formula [29], which expresses $L_p'(E,0)$ through the p-adic height of a Heegner point; combined with the complex formula, it shows that the two orders of vanishing agree when one of them is 1. For $r \geq 2$ no such link is known.

12.6. **What is proved.** For each curve and each prime in the stated ranges, the method gives: $Ш(E/\mathbf{Q})[p^\infty]$ is trivial, the rank of $E(\mathbf{Q})$ equals the number of known independent points, the p-adic regulator is nonzero, and the main conjecture of Iwasawa theory holds for (E,p). In the language of (1), the p-parts of both sides of the conjectural formula are determined in the p-adic setting, and the algebraic side is computed exactly at p. These are the parts of the Birch–Swinnerton-Dyer formula that are accessible at present for curves of rank at least two.

Appendix A. The error of the Riemann sums

Proof of Proposition 2.3. For $b = \omega(a)(1+p)^j$ the coefficient $a_{n,k}$ of (5) is $\sum_b \mu(b+p^n\mathbf{Z}_p)\binom{j}{k}$, so

$$a_k - a_{n,k} = \sum_b \int_{b+p^n\mathbf{Z}_p} \left(\binom{\ell(x)}{k} - \binom{\ell(b)}{k} \right) d\mu(x).$$

Let $x \in b + p^n\mathbf{Z}_p$. Then $\omega(x) = \omega(b)$, and $\langle x \rangle = \langle b \rangle\, y$ with $y = x/b \in 1 + p^n\mathbf{Z}_p$. Since $\log_p(1+p^n\mathbf{Z}_p) = p^n\mathbf{Z}_p$ and $v_p(\log_p(1+p)) = 1$, we have $\ell(x) = j + t$ with $t = \ell(y) \in p^{n-1}\mathbf{Z}_p$. By Vandermonde's identity, valid for p-adic integers,

$$\binom{j+t}{k} - \binom{j}{k} = \sum_{i=1}^{k} \binom{j}{k-i}\binom{t}{i}, \qquad \binom{t}{i} = \frac{t}{i}\binom{t-1}{i-1}.$$

The binomial coefficients of p-adic integers are p-adic integers, so $v_p\binom{t}{i} \geq v_p(t) - v_p(i) \geq n - 1 - e_k$ for $1 \leq i \leq k$. Thus the integrand takes values in $p^{n-1-e_k}\mathbf{Z}_p$, and since μ is $\mathbf{Z}_p$-valued (Lemma 2.2), so does each integral. □

Appendix B. Two properties of modular symbols

Proof of Lemma 2.1(a) and (b). Since $f(\tau+1) = f(\tau)$, the substitution $\tau \mapsto \tau + 1$ gives $\lambda(r+1) = \lambda(r)$, hence $[r+1]^+ = [r]^+$; and $[-r]^+ = [r]^+$ is immediate from the definition.

For (b), the Hecke operator T_ℓ with $\ell \nmid N$ acts on weight-two forms by

$$(T_\ell f)(\tau) = \ell f(\ell\tau) + \frac{1}{\ell}\sum_{b=0}^{\ell-1} f\Big(\frac{\tau+b}{\ell}\Big),$$

and $T_\ell f = a_\ell f$. Integrate from r to $i\infty$. The substitution $w = \ell\tau$ gives $\int_r^{i\infty} \ell f(\ell\tau)\,d\tau = \int_{\ell r}^{i\infty} f(w)\,dw$, and $w = (\tau+b)/\ell$ gives $\int_r^{i\infty} \frac{1}{\ell} f(\frac{\tau+b}{\ell})\,d\tau = \int_{(r+b)/\ell}^{i\infty} f(w)\,dw$. Multiplying by $2\pi i$,

$$a_\ell \lambda(r) = \lambda(\ell r) + \sum_{b=0}^{\ell-1} \lambda\Big(\frac{r+b}{\ell}\Big).$$

Applying this to $-r$ and using $\lambda((-r+b)/\ell) = \lambda(-(r+b')/\ell)$ for $b' \equiv -b \pmod{\ell}$, which follows from periodicity, we may add the two identities and divide by $-2\Omega_E$. □

Appendix C. Subgroups of $\mathrm{GL}_2(\mathbf{F}_p)$

Proposition 6.1 uses the following form of Dickson's classification [10]; see also [35, §2]. We give a complete proof of the part that is needed. Throughout, $p \geq 5$; a *Cartan subgroup* of $\mathrm{GL}_2(\mathbf{F}_p)$ is the group of units of a semisimple commutative subalgebra of rank two of $M_2(\mathbf{F}_p)$, split if the algebra is $\mathbf{F}_p \times \mathbf{F}_p$ and nonsplit if it is $\mathbf{F}_{p^2}$.

Proposition C.1. *Let G be a subgroup of $\mathrm{GL}_2(\mathbf{F}_p)$ and H its image in $\mathrm{PGL}_2(\mathbf{F}_p)$. Then one of the following holds:*

(a) *$G \supseteq \mathrm{SL}_2(\mathbf{F}_p)$;*
(b) *G is contained in a Borel subgroup;*
(c) *G is contained in the normaliser of a split or nonsplit Cartan subgroup;*
(d) *every element of H has order at most 5.*

Proof. Case 1: p divides $\#G$. An element $g \in G$ of order p has both eigenvalues equal to 1, since they are pth roots of unity in a field of characteristic p, and $g \neq 1$; so $g-1$ is nilpotent of rank one and g fixes exactly one line, $\ker(g-1)$. A subgroup of order p is generated by any of its nontrivial elements, so it too fixes exactly one line. If G has a single subgroup of order p, it is normal, and G preserves its line: this is (b). Otherwise G contains two subgroups of order p fixing different lines. In a basis made of the two lines they are generated by $\left(\begin{smallmatrix}1 & a\\ 0 & 1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1 & 0\\ b & 1\end{smallmatrix}\right)$ with $ab \neq 0$, and conjugating by $\mathrm{diag}(a,1)$ we may take $a = 1$. Their powers give all matrices $\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1 & 0\\ y & 1\end{smallmatrix}\right)$ with $x, y \in \mathbf{F}_p$. These generate $\mathrm{SL}_2(\mathbf{F}_p)$. Given $M = \left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)$ of determinant one, multiplying on the right by $\left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$ makes α nonzero if it was zero; then multiplying on the left by $\left(\begin{smallmatrix}1 & 0\\ -\gamma/\alpha & 1\end{smallmatrix}\right)$ and on the right by $\left(\begin{smallmatrix}1 & -\beta/\alpha\\ 0 & 1\end{smallmatrix}\right)$ gives $\mathrm{diag}(\alpha, \alpha^{-1})$; and $\mathrm{diag}(t, t^{-1})$ is the product

$$\begin{pmatrix}1 & t\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ -t^{-1} & 1\end{pmatrix}\begin{pmatrix}1 & t\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & -1\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}1 & -1\\ 0 & 1\end{pmatrix}.$$

This is (a).

Case 2: p does not divide $\#G$. Then $\#H$ is prime to p. Let H act on $X = \mathbf{P}^1(\mathbf{F}_{p^2})$. A nontrivial $h \in H$ lifts to a non-scalar $g \in G$ of order prime to p; its minimal polynomial divides $X^m - 1$ with $p \nmid m$, which is separable, so g is semisimple with two distinct eigenvalues, both in $\mathbf{F}_{p^2}$. Hence h fixes exactly two points of X, the two eigenlines of g. The stabiliser in H of a point of X is the image of a subgroup of a Borel subgroup of $\mathrm{GL}_2(\mathbf{F}_{p^2})$ of order prime to p, hence of a torus, and it is cyclic.

Count the pairs (h, x) with $h \in H \smallsetminus \{1\}$ and $hx = x$. Each h contributes 2. On the other side, group the points with nontrivial stabiliser into H-orbits $O_1, \dots, O_k$, with stabilisers of order $e_i \geq 2$, so that $\#O_i = \#H/e_i$. Then

$$2(\#H - 1) = \sum_{i=1}^{k} \#O_i\,(e_i - 1), \qquad \text{that is,} \qquad 2 - \frac{2}{\#H} = \sum_{i=1}^{k}\Big(1 - \frac{1}{e_i}\Big).$$

If $H = 1$ then G consists of scalars and lies in every Cartan subgroup. Otherwise the left side lies in $[1, 2)$ and each term on the right in $[\frac{1}{2}, 1)$, so $k \in \{2, 3\}$.

If $k = 2$, then $\frac{1}{e_1} + \frac{1}{e_2} = \frac{2}{\#H}$ with $e_i \le \#H$ forces $e_1 = e_2 = \#H$: two points x, y are fixed by all of H. If $k = 3$, then $\sum \frac{1}{e_i} = 1 + \frac{2}{\#H} > 1$, and with $e_1 \le e_2 \le e_3$ the only solutions are $(2, 2, n)$ with $\#H = 2n$, and $(2, 3, 3)$, $(2, 3, 4)$, $(2, 3, 5)$. In the case $(2, 2, n)$ with $n \ge 3$, the orbit with stabiliser of order n has $\#H/n = 2$ points $\{x, y\}$, and H preserves this pair. In the remaining cases, $(2, 2, 2)$ and the three exceptional ones, every nontrivial element of H fixes a point and so lies in a stabiliser of order $e_i \le 5$; this is (d).

It remains to treat a pair $\{x, y\} \subset X$ fixed pointwise or preserved by H. Since H is defined over $\mathbf{F}_p$, so is the pair as a set: if $x, y \in \mathbf{P}^1(\mathbf{F}_p)$, the elements of $\mathrm{GL}_2(\mathbf{F}_p)$ fixing both lines form a split Cartan subgroup C, and those preserving the pair form its normaliser. If x and y are not rational, they are conjugate under $\mathrm{Gal}(\mathbf{F}_{p^2}/\mathbf{F}_p)$, and the elements of $\mathrm{GL}_2(\mathbf{F}_p)$ fixing both form a nonsplit Cartan subgroup: choosing v on the line x, the map $\mathbf{F}_{p^2} \to \mathbf{F}_{p^2}^2$, $\lambda \mapsto \lambda v + \bar{\lambda}\bar{v}$ identifies $\mathbf{F}_p^2$ with $\mathbf{F}_{p^2}$, and the stabiliser of both lines with multiplications by $\mathbf{F}_{p^2}^\times$. Again those preserving the pair form the normaliser. In both cases G lies in the normaliser, which is (c). □

Two facts about normalisers are used in Proposition 6.1. Let C be a Cartan subgroup with eigenlines x, y over $\mathbf{F}_{p^2}$, and $N(C)$ its normaliser. An element of $N(C) \smallsetminus C$ exchanges x and y, so in a basis of $\mathbf{F}_{p^2}^2$ made of vectors on x and y it is antidiagonal, and its trace is zero. A split Cartan subgroup fixes a rational line, so it lies in a Borel subgroup. Finally, an element of a nonsplit Cartan subgroup is multiplication by some $\lambda \in \mathbf{F}_{p^2}^\times$; it is scalar if $\lambda \in \mathbf{F}_p$, and otherwise its characteristic polynomial is the minimal polynomial of λ, which is irreducible.

Appendix D. The p-adic sigma function

This appendix makes the definition of §3 explicit. Let E be given by a Weierstrass equation with integral coefficients, minimal at $p \ge 5$, and recall

$$b_2 = a_1^2 + 4a_2, \quad b_4 = 2a_4 + a_1a_3, \quad b_6 = a_3^2 + 4a_6, \quad c_4 = b_2^2 - 24b_4, \quad c_6 = -b_2^3 + 36b_2b_4 - 216b_6.$$

D.1. **The formal group.** Put $t = -x/y$ and $w = -1/y$. Substituting $x = t/w$, $y = -1/w$ into the equation gives $w = t^3 + a_1tw + a_2t^2w + a_3w^2 + a_4tw^2 + a_6w^3$, and iterating this identity from $w = t^3$ determines a unique series $w(t) = t^3(1 + a_1t + \cdots) \in \mathbf{Z}[[t]]$ [36, Ch. IV, Prop. 1.1]. Hence $x = t/w$ and $y = -1/w$ are Laurent series

$$x = t^{-2} - a_1t^{-1} - a_2 - a_3t - \cdots, \qquad y = -t^{-3} + \cdots,$$

with integral coefficients, and $\omega_E = dx/(2y + a_1x + a_3) = \big(1 + a_1t + (a_1^2 + a_2)t^2 + \cdots\big)dt$ with coefficients in $\mathbf{Z}$ [36, Ch. IV, Prop. 1.1 and §4]. The formal logarithm

$$z(t) = \int \omega_E = t + \frac{a_1}{2}t^2 + \frac{a_1^2 + a_2}{3}t^3 + \cdots$$

has the coefficient of t^n in $n^{-1}\mathbf{Z}$, so it converges on $p\mathbf{Z}_p$ and $v_p(z(t)) = v_p(t)$ there. The inverse map of the formal group, $P \mapsto -P$, sends t to a series $\iota(t) = -t + \cdots$ with $z(\iota(t)) = -z(t)$, because ω_E is invariant and changes sign under $[-1]$. A function of t is called *odd* if $f(\iota(t)) = -f(t)$; in the variable z this is ordinary oddness.

Lemma D.1. *As a Laurent series in z,*

$$x + \frac{b_2}{12} = z^{-2} + \sum_{k \ge 1} \gamma_k z^{2k},$$

where

$$\gamma_1 = \frac{c_4}{240}, \qquad \gamma_2 = \frac{c_6}{6048}, \qquad \gamma_k = \frac{3}{(2k+3)(k-2)} \sum_{i+j=k-1} \gamma_i \gamma_j \quad (k \geq 3).$$

Proof. Write $\wp = x + b_2/12$ and $' = d/dz$. Since $dz = \omega_E$, we have $x' = 2y + a_1 x + a_3$, and completing the square in the Weierstrass equation gives $(2y+a_1x+a_3)^2 = 4x^3+b_2x^2+2b_4x+b_6$. Expanding $4\wp^3 - g_2\wp - g_3$ with $g_2 = c_4/12$, $g_3 = c_6/216$ and comparing coefficients of $x^3, x^2, x, 1$ shows that this is the same cubic, so

$$\wp'^2 = 4\wp^3 - g_2\wp - g_3.$$

The function x is invariant under $[-1]$, so $\wp$ is even in z. To find its constant term, invert $z(t)$: with $\alpha = a_1/2$ and $\beta = (a_1^2 + a_2)/3$ one has $t = z - \alpha z^2 + (2\alpha^2 - \beta)z^3 + O(z^4)$, hence

$$t^{-2} = z^{-2} + a_1 z^{-1} + \Big(\frac{5a_1^2}{12} + \frac{2a_2}{3}\Big) + O(z), \qquad -a_1 t^{-1} = -a_1 z^{-1} - \frac{a_1^2}{2} + O(z),$$

and the constant term of x is $\frac{5a_1^2}{12} + \frac{2a_2}{3} - \frac{a_1^2}{2} - a_2 = -\frac{b_2}{12}$. So $\wp = z^{-2} + \sum_{k\geq 1} \gamma_k z^{2k}$. Comparing the coefficients of z^{-2} and z^0 in the differential equation gives $-8\gamma_1 = 12\gamma_1 - g_2$ and $-16\gamma_2 = 12\gamma_2 - g_3$. Differentiating it and dividing by $2\wp'$ gives $\wp'' = 6\wp^2 - g_2/2$, whose coefficient of z^{2k-2} for $k \geq 3$ reads $2k(2k-1)\gamma_k = 12\gamma_k + 6\sum_{i+j=k-1}\gamma_i\gamma_j$, and $2k(2k-1) - 12 = 2(2k+3)(k-2)$. □

Proposition D.2. *For every $c \in \mathbf{Q}_p$ there is exactly one odd series $\sigma \in t + t^2\mathbf{Q}_p[[t]]$ with*

$$x + c = -\frac{d}{\omega_E}\Big(\frac{1}{\sigma}\frac{d\sigma}{\omega_E}\Big),$$

namely

$$\sigma(t) = z \exp\Big(\frac{b_2 - 12c}{24}\, z^2 - \sum_{k\geq 1} \frac{\gamma_k}{(2k+1)(2k+2)}\, z^{2k+2}\Big), \qquad z = z(t).$$

Proof. In the variable z the equation reads $-(\log\sigma)'' = \wp + c - b_2/12$. Any solution of the form $z + O(z^2)$ satisfies $\log(\sigma/z) \in z\mathbf{Q}_p[[z]]$, and integrating twice with Lemma D.1,

$$\log\frac{\sigma}{z} = \frac{b_2 - 12c}{24} z^2 - \sum_{k\geq 1} \frac{\gamma_k}{(2k+1)(2k+2)} z^{2k+2} + Az$$

for a constant A; here the term $-\log z$ coming from z^{-2} is absorbed by the factor z. The right-hand side without Az is even in z, so σ is odd exactly when $A = 0$. Conversely the displayed series is odd, begins with $z = t + \cdots$, and satisfies the equation. □

So the choice of c is the only freedom, and the theorem of Mazur and Tate is the statement that exactly one value of c makes all coefficients of σ in t integral.

Theorem D.3 (Mazur–Tate [26])**.** *Let $p \geq 5$ be good ordinary. There is exactly one $c \in \mathbf{Z}_p$ for which the series of Proposition D.2 lies in $\mathbf{Z}_p[[t]]$. It is the slope of the unit-root eigenline of Frobenius on $H^1_{\mathrm{dR}}(E/\mathbf{Q}_p)$: if $\eta = x\omega_E$, then $\eta + c\,\omega_E$ is an eigenvector of the Frobenius φ with unit eigenvalue α. Equivalently $c = (b_2 - e_2)/12$ with $e_2 = E_2(E, \omega_E)$.*

The second description is due to Katz [17], and it is the one used for computation: the matrix of φ in the basis ω_E, η is computed to any p-adic precision by Kedlaya's algorithm [18], and since the second eigenvalue p/α has positive valuation, the columns of φ^m converge to multiples of the unit-root eigenvector [25, §3].

Lemma D.4. *Let σ be the canonical sigma function and $t_0 \in p\mathbf{Z}_p$. If σ is known modulo t^{M+1} with coefficients modulo p^N, then $\sigma(t_0)/t_0$ is known modulo $p^{\min(N,M)}$.*

Proof. The coefficient s_k of t^k is a p-adic integer, so $s_k t_0^{k-1} \in p^{k-1}\mathbf{Z}_p$; the omitted terms have $k \geq M+1$, and the error in each s_k with $k \leq M$ is multiplied by $t_0^{k-1} \in \mathbf{Z}_p$. $\square$

D.2. **An example.** For E_4 and $p = 5$ we have $b_2 = -3$, $c_4 = 3801$, $c_6 = -232\,605$, and Lemma D.1 gives

$$x - \tfrac{1}{4} = z^{-2} + \tfrac{1267}{80} z^2 - \tfrac{8615}{224} z^4 + \tfrac{1605289}{19200} z^6 + O(z^8);$$

the third coefficient is $\gamma_1^2/3$, as the recursion predicts. The matrix of Frobenius, with the images of ω_E and η as columns, is

$$\begin{pmatrix} 4\cdot 5 + 3\cdot 5^2 & 1 + 2\cdot 5 + 3\cdot 5^2 \\ 3\cdot 5 + 2\cdot 5^2 & 1 + 5^2 \end{pmatrix} + O(5^3),$$

of trace $1 + 4\cdot 5 + 4\cdot 5^2 \equiv a_5 = -4 \pmod{5^3}$. Its unit-root eigenvector is $\eta + c\,\omega_E$ with

$$c = 1 + 3\cdot 5 + 2\cdot 5^2 + 4\cdot 5^3 + O(5^4), \qquad \alpha = 1 + 3\cdot 5 + 2\cdot 5^2 + 3\cdot 5^3 + O(5^4),$$

so $e_2 = b_2 - 12c = 5 + 3\cdot 5^2 + 0\cdot 5^3 + O(5^4)$. Proposition D.2 then gives

$$\sigma_5(t) = t + \tfrac{1}{2}t^2 + (4\cdot 5 + 5^2)t^3 + (1 + 2\cdot 5 + 5^2)t^4 + (3 + 2\cdot 5 + 3\cdot 5^2)t^5 + (3 + 5 + 4\cdot 5^2)t^6 + \cdots,$$

each coefficient modulo 5^3. The coefficient of t^3 is $(a_1^2 + a_2)/3 + e_2/24 = e_2/24$, which is a 5-adic integer for every $e_2 \in \mathbf{Z}_5$; integrality constrains c only through the higher coefficients, which by Theorem D.3 are all integral for the value of c above and for no other. With $m = 10$, the point $10P_1$ has $v_5(t) = 1$ and $v_5(e(10P_1)) = 1$, and

$$\hat{h}_5(P_1) = \frac{2}{100} \log_5 \frac{e(10P_1)}{\sigma_5(10P_1)} = 2\cdot 5^{-1} + 1 + 5 + 2\cdot 5^2 + 2\cdot 5^3 + O(5^4),$$

which is the entry $\langle P_1, P_1 \rangle_5$ quoted in §3.1.

Appendix E. Saturation for the rank-four curve at five

For $\ell \in \{13, 23, 59, 67\}$ the groups $\tilde{E}_4(\mathbf{F}_\ell)$ are isomorphic to $\mathbf{Z}/20$, $\mathbf{Z}/30$, $\mathbf{Z}/30 \times \mathbf{Z}/2$ and $\mathbf{Z}/75$, of orders $m_\ell = 20$, 30, 60, 75. In each case the 5-Sylow subgroup is cyclic, so $R \mapsto (m_\ell/5)R$ induces an isomorphism of $\tilde{E}_4(\mathbf{F}_\ell)/5\tilde{E}_4(\mathbf{F}_\ell)$ onto the subgroup of order 5. That subgroup is generated by $g_\ell = (m_\ell/5)\tilde{P}_1$, which is

$$g_{13} = (8, 7), \qquad g_{23} = (4, 16), \qquad g_{59} = (56, 56), \qquad g_{67} = (27, 51),$$

and we let $\phi_\ell(R) \in \mathbf{F}_5$ be the discrete logarithm of $(m_\ell/5)\tilde{R}$ to the base g_ℓ. Computing the multiples $k g_\ell$ for $k \leq 4$ and comparing them with $(m_\ell/5)\tilde{P}_i$ gives the rows $(\phi_\ell(\tilde{P}_1), \ldots, \phi_\ell(\tilde{P}_4))$:

$$\begin{pmatrix} 1 & 3 & 2 & 4 \\ 1 & 4 & 1 & 2 \\ 1 & 4 & 0 & 1 \\ 1 & 4 & 1 & 3 \end{pmatrix} \qquad (\ell = 13,\ 23,\ 59,\ 67),$$

and the 4×4 matrix they form has determinant -1, a unit modulo 5. By Proposition 7.1, $P_1, \ldots, P_4$ are 5-saturated. The same construction, with the auxiliary primes listed in the data archive, gives saturation for every pair in Theorems A–D.

Appendix F. Tables

TABLE 2. E_4: for each good ordinary prime $5 \le p < 1000$, the residue of a_4 modulo p (modulo p^2 for $p = 7, 157$) and the surjectivity witness (q_1, q_2, q_3) of Proposition 6.1. In every case $v_p(S_p) = 0$ except at 7 and 157, where $v_p(S_p) = 1$.

p	a_4	witness	p	a_4	witness	p	a_4	witness	p	a_4	witness
5	1	3,13,3	7	$3 \cdot 7$	5,3,3	11	2	3,17,3	13	11	11,3,3
17	4	3,5,3	19	1	5,3,3	23	12	3,29,3	29	2	3,5,3
31	23	5,3,3	37	32	11,3,3	41	1	3,5,3	43	33	5,3,3
47	31	3,29,3	53	25	3,5,3	61	14	11,3,3	67	46	5,3,3
71	22	3,23,3	73	14	23,3,3	79	74	5,3,3	83	4	3,11,3
89	86	3,5,3	97	73	23,3,3	103	66	5,3,3	107	70	3,11,3
109	15	11,3,3	113	21	3,5,3	127	123	5,3,3	131	109	3,11,3
137	11	3,5,3	139	121	5,3,3	149	15	3,5,3	151	83	5,3,3
157	$34 \cdot 157$	11,3,3	163	14	5,3,3	167	121	3,29,3	173	129	3,5,3
179	136	3,11,3	181	157	11,3,3	191	8	3,23,3	193	90	29,3,3
197	77	3,5,3	199	98	5,3,3	211	210	5,3,3	223	104	5,3,3
227	120	3,11,3	229	30	11,3,3	233	9	3,5,3	239	41	3,23,3
241	153	23,3,3	251	93	3,11,3	257	128	3,5,3	263	135	3,23,3
269	38	3,5,3	271	131	5,3,3	277	8	11,3,3	281	249	3,5,3
283	81	5,3,3	293	121	3,5,3	307	242	5,3,3	311	260	3,47,3
313	133	23,3,3	317	21	3,5,3	331	98	5,3,3	347	149	3,11,3
349	211	11,3,3	353	305	3,5,3	359	19	3,23,3	367	308	5,3,3
373	207	11,3,3	379	162	5,3,3	383	211	3,29,3	389	325	3,5,3
397	267	11,3,3	401	307	3,5,3	409	232	23,3,3	419	347	3,11,3
421	14	11,3,3	431	227	3,23,3	433	149	23,3,3	439	65	5,3,3
443	18	3,11,3	449	64	3,5,3	457	452	29,3,3	461	73	3,5,3
463	429	5,3,3	467	264	3,11,3	479	294	3,41,3	487	415	5,3,3
491	269	3,11,3	499	21	5,3,3	503	72	3,29,3	509	29	3,5,3
521	351	3,5,3	523	344	5,3,3	541	495	11,3,3	547	278	5,3,3
557	179	3,5,3	563	161	3,11,3	569	288	3,5,3	571	406	5,3,3
577	121	23,3,3	587	578	3,11,3	593	515	3,5,3	599	214	3,23,3
601	247	23,3,3	607	512	5,3,3	613	46	11,3,3	617	333	3,5,3
619	320	5,3,3	631	320	5,3,3	641	199	3,5,3	643	37	5,3,3
647	640	3,29,3	653	210	3,5,3	659	153	3,11,3	661	513	11,3,3
673	152	29,3,3	677	358	3,5,3	683	56	3,11,3	691	18	5,3,3
701	236	3,5,3	709	357	11,3,3	719	124	3,67,3	727	521	5,3,3
733	479	11,3,3	739	364	5,3,3	743	740	3,23,3	751	333	5,3,3
757	562	11,3,3	761	424	3,5,3	769	735	23,3,3	773	215	3,5,3
787	493	5,3,3	797	748	3,5,3	809	488	3,5,3	811	782	5,3,3
821	642	3,5,3	823	286	5,3,3	827	283	3,11,3	829	691	11,3,3
839	533	3,41,3	853	744	11,3,3	857	490	3,5,3	859	64	5,3,3
863	796	3,23,3	877	221	11,3,3	881	111	3,5,3	883	840	5,3,3
887	788	3,29,3	907	304	5,3,3	911	857	3,23,3	919	104	5,3,3
929	720	3,5,3	937	418	23,3,3	941	346	3,5,3	947	913	3,11,3
953	587	3,5,3	967	499	5,3,3	971	691	3,11,3	977	113	3,5,3
983	863	3,29,3	991	657	5,3,3	997	202	11,3,3			

TABLE 3. The 45 rank-three pairs with $v_p(S_p) \geq 2$, with a_3 modulo $p^{v_p(S_p)+1}$. In each case $v_p(a_3) = v_p(S_p)$ and the leading digit of a_3 is that of S_p.

curve	p	a_3	curve	p	a_3	curve	p	a_3
43556b1	5	5^2	45352a1	5	$3 \cdot 5^2$	47087a1	5	$4 \cdot 5^2$
47696h1	5	$2 \cdot 5^2$	49411a1	5	$3 \cdot 5^2$	53122a1	5	5^3
53957b1	5	$2 \cdot 5^2$	63574d1	5	$2 \cdot 5^2$	74216a1	5	$4 \cdot 5^2$
74434a1	5	$2 \cdot 5^2$	80256a1	5	$4 \cdot 5^2$	83928b1	5	$3 \cdot 5^2$
87866a1	5	5^2	88502f1	5	$3 \cdot 5^2$	91506a1	5	$2 \cdot 5^2$
91958a1	5	$4 \cdot 5^2$	94281b1	5	$2 \cdot 5^2$	38600m1	7	7^2
49626a1	7	$4 \cdot 7^2$	51052a1	7	$4 \cdot 7^2$	53122a1	7	$3 \cdot 7^2$
54489b1	7	$4 \cdot 7^2$	59450i1	7	$5 \cdot 7^2$	61886d1	7	$4 \cdot 7^2$
65122e1	7	$5 \cdot 7^2$	70449b1	7	$6 \cdot 7^2$	71688a1	7	7^2
73931a1	7	$5 \cdot 7^2$	74864b1	7	$2 \cdot 7^2$	78287b1	7	$2 \cdot 7^2$
81328c1	7	$4 \cdot 7^2$	88022a1	7	$4 \cdot 7^2$	88883a1	7	$5 \cdot 7^2$
90953a1	7	7^3	33609a1	11	$10 \cdot 11^2$	67064a1	11	$6 \cdot 11^2$
89294a1	11	$5 \cdot 11^3$	92651a1	11	$5 \cdot 11^2$	62576b1	13	$12 \cdot 13^2$
97111a1	13	$2 \cdot 13^2$	98231a1	13	$6 \cdot 13^2$	88022a1	17	$2 \cdot 17^2$
61651a1	19	$6 \cdot 19^2$	36739a1	29	$17 \cdot 29^2$	60803a1	89	$84 \cdot 89^2$

TABLE 4. The rank-three pairs of Theorem B by prime: the number of pairs, their distribution by $v = v_p(S_p)$, the number with $a_p \equiv 1 \pmod{p}$ (anomalous), and p times the proportion of irregular pairs. The level at which a pair is settled is $v + 2$.

p	pairs	$v = 0$	$v = 1$	$v = 2$	$v = 3$	$a_p \equiv 1$	p·irr.
5	406	327	62	16	1	132	0.97
7	435	374	44	16	1	1	0.98
11	429	393	32	3	1	2	0.92
13	447	415	29	3	0	6	0.93
17	447	415	31	1	0	6	1.22
19	450	427	22	1	0	3	0.97
23	443	426	17	0	0	8	0.88
29	446	429	16	1	0	10	1.11
31	451	439	12	0	0	9	0.82
37	476	465	11	0	0	15	0.86
41	448	436	12	0	0	6	1.10
43	451	440	11	0	0	22	1.05
47	440	433	7	0	0	7	0.75
53	469	457	12	0	0	8	1.36
59	449	441	8	0	0	5	1.05
61	468	457	11	0	0	18	1.43
67	478	472	6	0	0	7	0.84
71	442	436	6	0	0	11	0.96
73	473	467	6	0	0	6	0.93
79	463	458	5	0	0	17	0.85
83	450	447	3	0	0	0	0.55
89	458	450	7	1	0	13	1.55
97	485	481	4	0	0	8	0.80
all	10404	9985	374	42	3	320	

Declarations

Data availability. The modular symbols, regulators, surjectivity witnesses and auxiliary primes for every pair, the programs that produced them, and Lean 4 files that recheck the surjectivity witnesses, the point counts and the points, are archived in [42]. The values were computed with PARI/GP 2.17.4, Cremona's eclib, and the modular-symbol and p-adic routines of SageMath 10.8.

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Use of AI tools. Claude Code (Anthropic) was used to write and run the programs behind the computations and to prepare the manuscript; the author checked every statement and is responsible for the content.

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Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India

Formerly, University of Calcutta (Heramba Chandra College), 23/49 Gariahat Road, Golpark, Kolkata 700029, West Bengal, India

Email address: d.bhattacharjee@erl-forschung.de
Email address: itsdeep@live.com
Email address: egspl@proton.me (GitHub-related contact; GitHub account creelie)
URL: https://orcid.org/0000-0003-0466-750X